United States Patent Application Publication 20250286582
Kind Code A1
Publication Date September 11, 2025
Inventor(s) Rupe; Jason W.

METHODS AND SYSTEMS FOR IDENTIFYING A FEATURE IN A COMMUNICATION NETWORK

Abstract

A method for identifying a feature in a communication network includes (a) receiving, at a first termination device of the communication network, a message specifying an assignment of first network resources of the communication network to the first termination device, (b) sending, from the first termination device, one or more transmitted signals into infrastructure of the communication network at one or more times specified by the assignment of the first network resources, the one or more transmitted signals being within one or more frequency ranges specified by the assignment of the first network resources, and (c) capturing, at the first termination device, one or more reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by one or more features of the infrastructure of the communication network.

Inventors: **Rupe; Jason W. (Lafayette, CO)**

Applicant: **Cable Television Laboratories, Inc.** (Louisville, CO)

Family ID: **96949660**

Appl. No.: **19/072217**

Filed: **March 06, 2025**

Related U.S. Application Data

us-provisional-application US 63561847 20240306

Publication Classification

Int. Cl.: **H04B3/46** (20150101); **H04L5/00** (20060101)

U.S. Cl.:

CPC **H04B3/46** (20130101); **H04L5/0048** (20130101);

Background/Summary

RELATED APPLICATIONS

[0001] This application claims benefit of U.S. Provisional Patent Application No. 63/561,847, filed on Mar. 6, 2024, which is incorporated herein by reference.

BACKGROUND

[0002] Communication mediums are used to transmit data in a communication network. Examples of communication mediums include, but are not limited to, coaxial electrical cables, twisted pair electrical cables (e.g., Ethernet cables and telephone cables), optical cables, and free space (e.g., for free space radio wireless communication links and free space optical wireless communication links). Communication networks are increasingly capable of transmitting data though a communication medium at a high bandwidth, such as by using sophisticated modulation techniques. However, these modulation techniques frequently require a substantially ideal communication medium, and an impairment in the communication medium, such as an impedance discontinuity, may therefore significantly degrade, or even completely inhibit, data transmission through the communication medium. Therefore, it is desirable to identify an impairment in a communication medium, such as to enable the impairment to be mitigated.

SUMMARY

[0003] Disclosed herein are methods and systems for identifying a feature, such as an impairment, in a communication network using one or more termination devices of the communication network. In certain embodiments, a termination device sends one or more transmitted signals into communication infrastructure of a communication network, and the termination device subsequently captures one or more reflection signals from the communication infrastructure resulting from the reflection of the one or more transmitted signals by one or more features in the communication network. The reflection signals are processed, such as using TDR and/or other processing techniques, to determine the location of the features relative to the termination device and/or to determine other characteristics, such as reflection strength, of the features. Additionally, in some embodiments, a termination device is configured to effectively reflect a signal received from communication infrastructure of a communication network back into the communication infrastructure as a loopback signal, to facilitate identification of features in the communication network. Furthermore, in certain embodiments, reflection signals are captured by two or more termination devices of a communication network, and a feature of the communication network is identified at least partially based on respective reflection signals received by two or more of the termination devices. Moreover, particular embodiments leverage phase information in a communication network to determine relative location of one of more features from a termination device of the communication network.

[0004] In an embodiment, a method for identifying a feature in a communication network includes (a) receiving, at a first termination device of the communication network, a message specifying an assignment of first network resources of the communication network to the first termination device, (b) sending, from the first termination device, one or more transmitted signals into infrastructure of the communication network at one or more times specified by the assignment of the first network resources, the one or more transmitted signals being within one or more frequency ranges specified by the assignment of the first network resources, and (c) capturing, at the first termination device, one or more reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by one or more features of the infrastructure in the communication network.

[0005] In an embodiment, a method for identifying a feature in a communication network includes (a) receiving, at a first termination device of the communication network, a message instructing the first termination device to monitor infrastructure of the communication network for presence of signals within a specified time period and within a specified frequency range, (b) capturing, at the first termination device, a transmitted signal during the specified time period, (c) sending, from the first termination device, a loopback signal into the infrastructure of the communication network, the loopback signal being a function of the transmitted signal captured by the first termination device, and (d) capturing, at the first termination device, one or more first reflection signals from the infrastructure of the communication network resulting from reflection of the loopback signal by one or more features of the infrastructure of the communication network.
[0006] In an embodiment, a method for identifying a feature in a communication network includes (a) receiving, at a first termination device of the communication network, a message instructing the first termination device to monitor infrastructure of the communication network for presence of signals within a specified time period and within a specified frequency range, (b) receiving, at a second termination device of the communication network, a message instructing the second termination device to monitor the infrastructure of the communication network for presence of signals within the specified time period and within the specified frequency range, (c) capturing, at the first termination device, one or more first reflection signals from the infrastructure of the communication network resulting from reflection of one or more transmitted signals by one or more first features in the infrastructure of the communication network, the one or more transmitted signals being within the specified frequency range, and (d) capturing, at the second termination device, one or more second reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by the one or more first features in the infrastructure of the communication network.
[0007] In an embodiment, a method for determining respective locations of one or more features of a communication network relative to a cable modem of the communication network includes (a) controlling the cable modem and a network hub of the communication network to determine a channel response of the communication network between the cable modem and the network hub using one or more of (i) a channel estimation process, (ii) a pre-equalization process, and (iii) a symbol capture process, (b) converting the channel response from a frequency domain to a time domain, (c) determining signal phase information from the channel response in the time domain, and (d) estimating respective distances of the one or more features of the communication network from the cable modem at least partially based on the signal phase information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is a schematic diagram of a communication environment including a system for identifying a feature in a communication network, according to an embodiment.
[0009] FIG. **2** is a flow chart of a method for identifying a feature in a communication network, according to an embodiment.
[0010] FIG. **3** is a schematic diagram of the FIG. **1** communication environment illustrating two example impairments in communication infrastructure.
[0011] FIG. **4** is a signal flow diagram illustrating examples of signal flows within a communication network of the FIG. **1** communication environment while executing the FIG. **2** method.
[0012] FIG. **5** is a flow chart of another method for identifying a feature in a communication network, according to an embodiment.
[0013] FIG. **6** is a signal flow diagram illustrating examples of signal flows within the

communication network of the FIG. **1** communication environment while executing the FIG. **5** method.

[0014] FIG. **7** is a flow chart of an additional method for identifying a feature in a communication network, according to an embodiment.

[0015] FIG. **8** is a flow chart of another method for identifying a feature in a communication network, according to an embodiment.

[0016] FIG. **9** is a signal flow diagram illustrating examples of signal flows within the communication network of the FIG. **1** communication environment while executing the FIG. **8** method.

[0017] FIG. **10** is a flow chart of an additional method for identifying a feature in a communication network, according to an embodiment.

[0018] FIG. **11** is a signal flow diagram illustrating examples of signal flows within the communication network of the FIG. **1** communication environment while executing the FIG. **10** method.

[0019] FIG. **12** is a schematic diagram of an embodiment of the FIG. **1** communication environment where the communication network is embodied by a cable communication network.

[0020] FIG. **13** is a flow chart of a method for determining respective locations of one or more features of a communication network relative to a cable modem of the communication network, according to an embodiment.

[0021] FIG. **14** is a schematic diagram of an embodiment of the FIG. **1** communication environment where the communication network is embodied by a passive optical network (PON) communication network.

[0022] FIG. **15** is a schematic diagram of an embodiment of the FIG. **1** communication environment where the communication network is embodied by a coherent optical communication network.

[0023] FIG. **16** is a schematic diagram of an embodiment of the FIG. **1** communication environment where the communication network is embodied by a wireless communication network.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0024] Time Domain Reflectometry (TDR) may be used to determine one or more characteristics of an impairment in a communication medium. TDR operates on the principle that a portion of a signal traveling through a communication medium will be returned to its source as a reflection signal in response to the signal encountering a discontinuity in the communication medium. In this document, a “discontinuity” includes, for example, one or more of an impedance change, a change in index of refraction, an obstruction, or a reflective feature, such as a reflective surface. Location of the discontinuity may be determined from timing of the reflection signal, and characteristics of the discontinuity may be determined from characteristics of the reflection signal. For example, consider a cable communication network including a cable modem connected to a coaxial electrical cable where there is an impedance discontinuity in the coaxial electrical cable. A portion of a communication signal sent from the cable modem into the coaxial electrical cable will return to the cable modem as a reflected signal due to the communication signal encountering the impedance discontinuity. A distance between the cable modem and the impedance discontinuity along the coaxial electrical cable can be determined using TDR by dividing (i) a difference between a time when the reflection signal was received by the cable modem and a time that the communication signal was sent by the cable modem into the coaxial electrical cable by (ii) velocity of signal propagation along the coaxial electrical cable.

[0025] It has historically been difficult to apply TDR to a cable communication network due to the presence of filters and other devices which block the transmission of signals outside of specified frequency ranges in the cable communication network. However, such filters and other devices are increasingly absent from cable communication networks to enable use of new communication techniques, such as Full Duplex (FDX) communication according to a Data Over Cable Service Interface Specification (DOCSIS) 3.1 or 4.0 communication standard. As such, modern cable communication networks are increasingly capable of transmitting signals in frequency ranges which enable TDR more broadly. Applicant has determined that this capability of modern cable communication networks can be leveraged to perform TDR in a cable communication network using termination devices, i.e., cable modems, of the cable communication network.

[0026] Disclosed herein are methods and systems for identifying an impairment or other feature in a communication network using one or more termination devices of the communication network. In certain embodiments, a termination device sends one or more transmitted signals into the communication infrastructure of a communication network, and the termination device subsequently captures one or more reflection signals from the communication infrastructure resulting from reflection of the one or more transmitted signals by one or more features in the communication network. The reflection signals are processed, such as using TDR and/or other processing techniques, to determine location of the features relative to the termination device and/or to determine other characteristics, such as reflection strength, of the features. Additionally, in some embodiments, a termination device is configured to effectively reflect a signal received from the communication infrastructure of a communication network back into the communication infrastructure as a loopback signal, to facilitate identification of impairments or other features in the communication network. Furthermore, in certain embodiments, reflection signals are captured by two or more termination devices of a communication network, and an impairment or other feature of the communication network is identified at least partially based on respective reflection signals received by two or more of the termination devices. Moreover, particular embodiments leverage phase information in a communication network to determine relative location of one of more impairments or other features from a termination device of the communication network.

[0027] The new methods and systems are discussed below primarily with respect to a cable communication network application where the termination devices are cable modems and a network hub includes one or more of a cable modem termination system (CMTS) and a remote physical layer (R-PHY) device. However, it should be understood that the new methods and systems are not limited to a cable communication network application. For example, certain embodiments of the new methods and systems could be applied to a passive optical network (PON) communication network, a digital subscriber line (DSL) communication network, a coherent optical communication network, or a wireless communication network.

[0028] FIG. **1** is a schematic diagram of a communication environment **100** including a communication network **102** and external network resources **104**. As discussed below, communication network **102** includes elements which implement embodiments of the new methods and systems for identifying a feature in a communication network. Communication network **102** includes a network hub **106**, termination devices **108**, and communication infrastructure **110**. In this document, specific instances of an item may be referred to by use of a numeral in parentheses (e.g. termination device **108**(**1**)) while numerals without parentheses refer to any such item (e.g. termination devices **108**). Although FIG. **1** depicts communication network **102** as including four termination devices **108**, the quantity of termination devices **108** in communication network **102** may vary. For example, some embodiments of communication network **102** include only a single termination device **108**, while other embodiments of communication network **102** include a large quantity of termination devices **108**, such as tens, hundreds, thousands, or even more, termination devices **108**.

[0029] Each termination device **108** is configured to provide a respective interface to communication network **102**. For example, in some embodiments, one or more termination device **108** is a cable modem, a DSL modem, an optical network termination (ONT), an optical network unit (ONU), or a wireless modem. Each termination device **108** need not have the same configuration. In certain embodiments, one or more termination devices **108** are located at respective subscriber premises, such as to enable subscribers to communicatively interface user equipment, such as computers, mobile phones, televisions, Internet of things (IoT) devices, security devices, entertainment devices, etc., to communication networks **102**. Additionally, termination devices **108** are optionally integrated with one or more other devices. For example, in some embodiments, one or more termination devices **108** are integrated with a wireless access point and a network switch as part of a premises gateway. Furthermore, in particular embodiments, one or more of termination devices **108** are portable devices that may be easily moved within communication network **102**, such as to facilitate feature identification.

[0030] Communication infrastructure **110** communicatively couples each termination devices **108** to network hub **106**. Communication infrastructure **110** includes, for example, one or more coaxial electrical cables, optical cables, twisted pair electrical cables, coupling devices (e.g., taps, splices, and/or splitters), filters, domain conversion devices (e.g., fiber nodes and/or remote terminals), amplifiers, power inserters, wireless communication devices, etc. Although communication network **102** is illustrated as having a point to multi-point mesh topology, the topology of communication network **102** could vary. For example, in some embodiments, communication infrastructure **110** is configured as a ring such that communication network **102** has a ring topology. As another example, in certain embodiments, communication infrastructure **110** includes a respective communication medium for each termination device **108**, such that communication network **102** has a star topology. As an additional example, communication infrastructure **110** could be modified to have a mesh topology different from that of FIG. **1**.

[0031] Network hub **106** is configured to control communication between termination devices **108** and network hub **106** via communication infrastructure **110**, and network hub **106** is also configured to interface communication network **102** with external network resources **104**. Certain embodiments of network hub **106** are further configured to control communication between two or more termination devices **108** via communication infrastructure **110**, such as to enable peer-to-peer communication between termination devices **108**. In some embodiments, network hub **106** includes one or more of a CTMS, a R-PHY device, an optical line terminal (OLT), a broadband network gateway (BNG), an ONU, an ONT, a digital subscriber line access multiplexer (DSLAM), and a core wireless communication network. In particular embodiments, external network resources **104** include one or more of the Internet, an intranet, a content delivery system, a computing system (e.g., a distributed or “cloud” computing system), a conferencing system, a data storage system, a communication system, an authentication, authorization, and accounting system, an another communication network (e.g., a wide area network (WAN), a metropolitan area network (MAN), a local area network (LAN), and/or a transmission communication network).

[0032] Particular embodiments of communication network **102** are configured as an access communication network. For example, some embodiments of communication network **102** are configured as a cable communication network, such as discussed below with respect to FIG. **12**, a passive optical network (PON) communication network, such as discussed below with respect to FIG. **14**, a DSL communication network, or a wireless communication network, such as discussed below with respect to FIG. **16**. However, communication network **102** could be configured as another type of network. For example, communication network **102** could be configured as a coherent optical network, such discussed below with respect to FIG. **15**. As another example, communication network **102** could be configured as a hyperscaler data center network, a LAN, a WAN, or a MAN.

[0033] Network hub **106** is illustrated as including a controller **112** which cooperates with termination devices **108** to implement one or more embodiments of the new methods for identifying a feature in a communication network. Accordingly, controller **112** and termination devices **108** collectively implement an embodiment of the new systems for identifying a feature in a communication network. Controller **112** is formed, for example, of digital electronic circuitry and/or analog electronic circuitry. For instance, some embodiments of controller **112** are implemented by one or more processors executing non-transitory instructions, such as software and/or firmware, stored in a data store, e.g., stored in one or more memories. While controller **112** is depicted as being a discrete element of network hub **106**, controller **112** could be integrated with one or more elements of network hub **106**. For example, in certain embodiments of network hub **106** including a CMTS, controller **112** is at least partially implemented by one or more processors of the CMTS executing firmware and/or software stored in a memory of the CMTS. Additionally, controller **112** could be external to network hub **106** without departing from the scope hereof. For example, in some alternate embodiments, controller **112** is implemented in external network resources **104**. As another example, in some other alternate embodiments, controller **112** is a stand-alone device communicatively coupled to communication infrastructure **110**. As an additional example, in certain additional alternate embodiments, controller **112** is implemented in one or more termination devices **108**. Furthermore, controller **112** could be distributed among two or more elements within communication networks **102** and/or external to communication networks **102**. For example, controller **112** could be split between network hub **106** and one or more termination devices **108**. As another example, controller **112** could be distributed among two or more termination devices **108**. As a further example, controller **112** could be distributed between network hub **106** and external network resources **104**.

[0034] In certain embodiments of communication network **102**, one or more termination devices **108** are capable of sending signals into communication infrastructure **110** in a particular frequency range and receiving signals from communication infrastructure **110** in at least the same frequency range. Applicant has determined that this feature can be leveraged to perform TDR, and/or another feature identification method in communication network **102**, using solely a single termination device **108** as both a transmitted signal source and a signal receiver. Accordingly, particular embodiments of communication network **102** are configured to use a single termination device **108** to send transmitted signals into communication infrastructure **110** and to capture reflection signals from communication infrastructure **110** resulting from the transmitted signals encountering a feature, such as an impairment, in communication infrastructure **110**. Communication network **102** and/or an external device subsequently analyze the captured reflection signals to identify the feature in communication infrastructure **110**. Additionally, in some embodiments, one or more termination devices **108** are configured to capture ingress signals on communication infrastructure **110**, such as due to a defect in communication infrastructure **110**.

[0035] Discussed below with respect to FIGS. **2-7** are several examples of how certain embodiments of communication network **102** identify a feature in communication infrastructure **110** using a single termination device **108** as both a transmitted signal source and a signal receiver. The example methods of FIGS. **2** and **4-7** are discussed below with respect to a scenario illustrated in FIG. **3** where there is an impairment in communication infrastructure **110** affecting solely termination device **108**(**1**) as well as an ingress point in communication infrastructure **110** potentially affecting all termination devices **108**. It is understood, though, that the methods of FIGS. **2** and **4-7** could be used to identify other impairments in communication infrastructure **110**. Additionally, the example methods of FIGS. **2** and **4-7** are discussed below assuming that termination device **108**(**1**) serves as a transmitted signal source and a signal receiver. It is understood, however, that the example methods of FIGS. **2** and **4-7** could be implemented with another termination device **108** of communication network **102**, e.g., termination device **108**(**2**), **108**(**3**), or **108**(**4**) serving as a transmitted signal source and a signal receiver.

[0036] FIG. **2** is a flow chart of method **200** for identifying a feature, such as an impairment, in a communication network, which is performed by certain embodiments of communication network **102**. In some embodiments, communication network **102** performs method **200** solely during one or more quiet periods of communication network **102**, where a quiet period is a period when termination devices **108** are not communicating with network hub **106** via communication infrastructure **110** using frequency ranges that are allocated for feature identification. Method **200** is discussed below with respect to FIG. **3** which illustrates the following example impairments in communication network **102**: (i) a discontinuity **302** that is upstream of termination device **108**(**1**) and only affects termination device **108**(**1**) and (ii) an ingress point **304** that allows ingress of signals **306** into communication infrastructure **110**. Discontinuity **302** is, for example, an impedance discontinuity in a communication medium of communication infrastructure **110**, such as an impedance discontinuity in a coaxial electrical cable, a tap, a power inserter, etc., in embodiments where communication network **102** is a cable communication network. As another example, discontinuity **302** could be a change in index of refraction of an optical communication path in embodiments of communication infrastructure **110** including an optical communication medium. As a further example, discontinuity **302** could be a signal fade in a wireless communication link, such as caused by an obstruction. Ingress point **304** is, for example, an undesired opening in communication infrastructure **110** allowing ingress of signals other than those intended to be carried by communication infrastructure **110**, such as signals **306**. Method **200** is further discussed below with respect to a signal flow diagram **400** of FIG. **4** which illustrates examples of signal flows within communication network **102** while executing method **200**. Signal flow diagram **400** includes vertical lines logically representing each of network hub **106** (including controller **112**), termination device **108**(**1**), discontinuity **302**, and ingress point **304**. Other elements of communication network **102** are not shown in FIG. **4** for illustrative clarity, and FIG. **4** should not be considered as being to scale. An arrow **401** in FIG. **4** represents increasing time in the direction of the arrow.

[0037] In a block **202** of method **200**, a termination device **108** is selected to perform a feature search in communication network **102** over a predetermined time duration and over a predetermined frequency range of signals in communication infrastructure **110**. The predetermined time duration is, for example, a predetermined amount of time or a period spanning a predetermined starting time and a predetermined ending time. In one example of block **202**, controller **112** selects termination device **108**(**1**) to perform a feature search in communication network **102** over a predetermined time duration and over a predetermined frequency range, such as based on a predetermined schedule of feature searches or in response to a signal indicating impaired communication between network hub **106** and termination device **108**(**1**). In another example of block **202**, termination device **108**(**1**) selects itself to perform a feature search over a predetermined time duration and over a predetermined frequency range, such as in response to a predetermined schedule or in response to termination device **108**(**1**) encountering communication errors when communicating with network hub **106** via communication infrastructure **110**.

[0038] In a block **204** of method **200**, the termination device **108** selected in block **202**, henceforth referred to as “the selected termination device **108**” when discussing method **200**, requests network resources of communication network **102** from controller **112** for the feature search. The requested network resources include, for example, one or more time periods and one or more frequency ranges of signals within communication infrastructure **110** for use by the selected termination device **108** when performing the feature search. For instance, in some embodiments where (i) communication network **102** is a cable communication network, (ii) termination devices **108** are cable modems, and (iii) network hub **106** includes, for example, a CMTS, R-PHY device, or a remote media control layer and remote physical layer (R-MAC-PHY) device, the requested network resources include a request from a cable modem to a controller of the CMTS or R-PHY device for an idle slot in the cable communication network, where the idle slot is a combination of a particular time period and one or more frequency bands that are free of communication signals in communication infrastructure **110**. In an alternate embodiment of block **204**, controller **112** informs the selected termination device **108** that the selected termination device **108** is to perform a feature search. In one example of block **204**, termination device **108**(**1**) sends a message **402** to controller **112** of network hub **106** via communication infrastructure **110**, or out-of-band with respect to communication infrastructure **110**, requesting network resources to perform a feature search, as illustrated in FIG. **4**. In another example of block **204**, network hub **106** sends a message **404** to termination device **108**(**1**) via communication infrastructure **110**, or out-of-band with respect to communication infrastructure **110**, informing termination device **108**(**1**) that it is to perform a feature search, as illustrated in FIG. **4**.

[0039] In a block **206** of method **200**, (i) controller **112** identifies network resources to assign to the selected termination device **108** to perform the feature search, (ii) controller **112** allocates the identified network resources to the selected termination device **108**, (iii) controller **112** informs the selected termination device **108** that it has been assigned the identified network resources, and (iv) controller **112** optionally informs other termination devices **108** of the upcoming feature search by the selected termination device **108**. The network resources allocated and assigned in block **206** include, for example, one or more time periods and one or more frequency ranges of signals in communication infrastructure **110**. In embodiments where communication network **102** is a cable communication network, the network resources assigned to the selected termination device **108** include, for example, one or more idle slots of the cable communication network. In one example of block **206**, (i) controller **112** identifies **406** network resources (NR) **408** of communication network **102** for use by termination device **108**(**1**) to perform a feature search, (ii) controller **112** allocates **410** network resources **408** to termination device **108**(**1**), such as by updating a network resource allocation record (not shown) of network hub **106**, and (iii) controller **112** sends a message **412** to termination device **108**(**1**) including assignment of network resources **408** to termination device **108**(**1**), as illustrated in FIG. **4**.

[0040] In a block **208** of method **200**, the selected termination device **108** sends one or more transmitted signals into communication infrastructure **110**(i) during one or more time periods allocated in the assigned network resources and (ii) within one or more frequency ranges allocated in the assigned network resources. The transmitted signals include, for example, calibrated pilot signals and/or communication traffic bearing signals. Additionally, the selected termination device **108** listens for signals, i.e., it monitors communication infrastructure **110** for presence of signals, that are within at least the one or more frequency ranges encompassing the transmitted signals, at least partially during the one or more time periods allocated in the assigned network resources. The one or more transmitted signals may be short duration burst signals. Additionally, the selected termination device **108** may send transmitted signals into communication infrastructure **110** over multiple frequencies, such as according to a pattern of increasing frequencies where transmitted signal frequency increases with increasing time, a pattern of decreasing frequencies where transmitted signal frequency decreases with increasing time, a random pattern where transmitted signal frequency randomly varies with time, or any other pattern where transmitted signal frequency is a function of time. The selected termination device **108** is optionally configured to send a plurality of transmitted signals, e.g., a plurality of calibrated pilot signals and/or communication traffic bearing signals, into communication infrastructure **110** over the time period allocated in the assigned network resources, for example, to help increase accuracy of the feature search, such as by enabling confirmation of reflection signals resulting from an impairment or other network feature, or by enabling comparison of multiple reflection signals resulting from an impairment or other network feature. In one example of block **208**, termination device **108**(**1**) sends a transmitted signal **414** into communication infrastructure **110**, as illustrated in FIG. **4**. A portion of transmitted signal **414** returns to termination device **108**(**1**) via communication infrastructure **110** as a reflection signal **416** in response to transmitted signal **414** encountering discontinuity **302**, as also illustrated in FIG. **4**. A portion **418** of transmitted signal **414** remaining after generation of reflection signal **416** at discontinuity **302** continues to network hub **106** (and other elements of communication network **102** not shown in FIG. **4**) via communication infrastructure **110**.

[0041] In a block **210** of method **200**, the selected termination device **108** monitors communication infrastructure **110** and captures any signals received by the selected termination device **108** during, and optionally after, the one or more times periods allocated in the assigned network resources. The signals captured by the selected termination device **108** in block **210** include, for example, (i) one or more reflection signals resulting from the transmitted signals sent in block **208** encountering one or more impairments or other features in communication infrastructure **110** and/or (ii) ingress of signals into communication infrastructure **110**, such as due to defects in communication infrastructure **110**. The selected termination device **108** monitors communication infrastructure **110** for signals at least within the one or more frequency ranges allocated in the assigned network resources, in block **210**. Additionally, in some embodiments, the selected termination device **108** further monitors communication infrastructure **110** for signals outside of the one or more frequency ranges allocated in the assigned network resources, such as to increase likelihood of detection of a non-linear feature and/or to enable detection of ingress of signals that are outside of the pilot signal frequency ranges. In embodiments where communication network **102** is a cable communication network, the signals captured in block **210** include, for example, radio frequency reflection signals resulting from a feature and/or undesired ingress of radio frequency signals into the cable communication network. In one example of block **210**, termination device **108**(**1**) monitors **420** communication infrastructure **110** during one or more times periods allocated in assigned network resources **408**, and termination device **108**(**1**) captures (i) reflection signal **416** and (ii) signals **306** from ingress point **304**, while termination device **108**(**1**) monitors **420** communication infrastructure **110**, as illustrated in FIG. **4**.

[0042] The signals captured in block **210** may be expressed by the selected termination device **108** in a variety of manners. For example, the captured signals may be expressed as spectrum data (e.g., magnitude as a function of frequency), complex data (e.g., I, Q values), and/or Received Modulation Error Ratio (RxMER) per subcarrier. Additionally, the selected termination device **108** optionally converts any signals captured in block **210** into a form that facilitates processing of the captured signals.

[0043] In a block **212** of method **200**, the signals captured in block **210** are analyzed to identify one or more impairments or other features of communication infrastructure **110**. The analysis of block **212** is performed, for example, by the selected termination device **108**, controller **112**, and/or one or more other elements that are within, or external to, communication network **102**. In one example of block **212**, termination device **108**(**1**) and/or controller **112** determine a location of discontinuity **302** with respect to termination device **108**(**1**) along communication infrastructure **110** at least partially using TDR. For instance, in these embodiments, termination device **108**(**1**) and/or controller **112** may divide (i) a difference between a time when termination device **108**(**1**) captures reflection signal **416** and a time when termination device **108**(**1**) sends transmitted signal **414** into communication infrastructure **110** by (ii) propagation velocity of signals through communication infrastructure **110**, to determine distance of discontinuity **302** from termination device **108**(**1**). As another example of block **212**, termination device **108**(**1**) and/or controller **112** may determine reflection strength of discontinuity **302** by comparing strength of captured reflection signal **416** to strength of transmitted signal **414**, where reflection strength of discontinuity **302** increases with decreasing difference between strength of captured reflection signal **416** and strength of pilot signal **414**.

[0044] As a further example of block **212**, termination device **108**(**1**) and/or controller **112** may determine one or more characteristics of discontinuity **302** by (i) converting reflection signal **416** from a time domain to a frequency domain, such as using a Fourier analysis, and (ii) matching a pattern of the frequency domain of reflection signal **416** to a pattern of a known frequency domain corresponding to a particular impairment characteristic. For instance, in embodiments where discontinuity **302** results from water intrusion into a cable, termination device **108**(**1**) and/or controller **112** may determine that discontinuity **302** is characteristic of a water intrusion by (i) converting reflection signal **416** from a time domain to a frequency domain and (ii) matching a pattern of the frequency domain of reflection signal **416** to a pattern of a known frequency domain corresponding to a water impairment.

[0045] Reflection signals from a non-linear feature, such as an impairment, may have a different frequency than a transmitted signal, e.g., a pilot signal, ranging signal, sounding signal, or communication bearing traffic signal encountering the non-linear feature. Accordingly, in particular embodiments, termination device **108**(**1**) and/or controller **112** determine that discontinuity **302** is a non-linear impairment by performing the following procedure: (i) comparing frequency of reflection signal **416** to frequency of transmitted signal **414**, (ii) determining a difference in time between when termination device **108**(**1**) sends transmitted signal **414** into communication infrastructure **110** and a time when termination device **108**(**1**) receives reflection signal **416**, and (ii) determining that discontinuity **302** is a non-linear impairment in response to frequency of reflection signal **416** being different from frequency of transmitted signal **414** by at least a first threshold amount and the determined difference in time being less than a second threshold amount. Additionally, difference in frequency of captured signals **306** from frequency of transmitted signal **414** may indicate that captured signals **306** are ingress signals. Accordingly, in some embodiments, termination device **108**(**1**) and/or controller **112** may determine that captured signals **306** represent ingress into communication infrastructure **110** by performing the following procedure: (i) comparing frequency of captured signals **306** to frequency of transmitted signal **414** and (ii) determining that captured signals **306** are ingress signals in response to frequency of captured signals **306** being different from frequency of transmitted signal **414** by at least a threshold amount. Certain embodiments of communication network **102** further condition determining that captured signals **306** are ingress signals on termination device **108**(**1**) receiving captured signals **306** at least a minimum threshold amount of time after termination device **108**(**1**) sends pilot signal **414** into communication infrastructure **110**, to prevent a reflection signal from a non-linear feature being misidentified as an ingress signal.

[0046] In embodiments where the selected termination device **108** sends multiple transmitted signals into communication infrastructure **110**, the selected termination device **108** may capture multiple corresponding reflection signals due to presence of an impairment or other feature in communication infrastructure **110**. In these embodiments, termination device **108**(**1**) and/or controller **112** are optionally further configured to consider multiple reflection signals in block **212**. For example, termination device **108**(**1**) and/or controller **112** may determine an average or other mathematical function of multiple captured reflection signals, such as to improve accuracy of analysis in block **212**. As another example, termination device **108**(**1**) and/or controller **112** may compare multiple instances of captured reflection signals to determine whether any of the captured reflection signals are outliers, and optionally discard any outlying captured reflection signals, to promote accurate analysis in block **212**. As a further example, termination device **108**(**1**) and/or controller **112** may select one or more captured reflection signals considered most likely to be accurate, such as by voting or by a consensus procedure, and optionally discard the remaining reflection signals when performing analysis in block **212**.

[0047] FIG. **5** is a flow chart of a method **500** for identifying a feature, such as an impairment, in a communication network which is performed by certain embodiments of communication network **102**. Method **500** is discussed below with respect to the example impairments of FIG. **3** as well as with respect to a signal flow diagram **600** of FIG. **6** which illustrates examples of signal flows while executing method **500**. Signal flow diagram **600** includes vertical lines logically representing each of network hub **106** (including controller **112**), termination device **108**(**1**), discontinuity **302**, and ingress point **304**. Other elements of communication network **102** are not shown in FIG. **6** for illustrative clarity, and FIG. **6** should not be considered as being to scale. An arrow line **601** in FIG. **4** represents increasing time in a direction of the arrow.

[0048] The method of FIG. **5** includes a ranging process and/or a sounding process. A ranging process is a process where a termination device **108** synchronizes itself with network hub **106** by adjusting communication signal timing and power levels. In particular, the termination device **108** sends a ranging request signal to network hub **106**, and network hub **106** responds by sending a ranging response signal adjusting the termination device **108**'s signal timing and signal power as needed. Each of a ranging request signal and a ranging response signal can be referred to as a ranging signal. A sounding process, in turn, is a process where a communication channel between a termination device **108** and network hub **106** is assessed and optimized. The termination device **108** sends a series of sounding transmitted signals, such as Orthogonal Upstream Data Channels (OUDC) signals, to network hub **106**. Network hub **106** then analyzes the sounding transmitted signals to evaluate channel quality, and network hub **106** provides a sounding feedback signal to the termination device **108** suggesting adjustments to parameters such as modulation schemes and signal power levels. Each of a sounding transmitted signal and a sounding feedback signal can be referred to as a sounding signal.

[0049] Referring to FIG. **5**, in a block **502** of method **500**, a termination device **108** is selected to perform a ranging and/or sounding process in communication network **102** over a predetermined time duration and over a predetermined frequency range of signals in communication infrastructure **110**. The predetermined time duration is, for example, a predetermined amount of time or a period spanning a predetermined starting time and a predetermined ending time. In one example of block **502**, controller **112** selects termination device **108**(**1**) to perform a ranging and/or sounding process in communication network **102** over a predetermined time duration and over a predetermined frequency range, such as based on a predetermined schedule or in response to a signal indicating impaired communication between network hub **106** and termination device **108**(**1**).

[0050] In a block **504** of method **500**, controller **112** informs the termination device **108** selected in block **502**, henceforth referred to as "the selected termination device **108**" when discussing method **500**, that the selected termination device **108** is to perform a ranging and/or sounding process. In one example of block **504**, network hub **106** sends termination device **108**(**1**) a message **602** informing termination device **108**(**1**) that the termination device **108**(**1**) is to perform a ranging and/or sounding process, as illustrated in FIG. **6**. In a block **506** of method **500**, (i) controller **112** identifies network resources to assign to the selected termination device **108** to perform the ranging and/or sounding process, (ii) controller **112** allocates the identified network resources to the selected termination device **108**, (iii) controller **112** informs the selected termination device **108** that it has been assigned the identified network resources, and (iv) controller **112** optionally informs other termination devices **108** of the upcoming ranging and/or sounding process by the selected termination device **108**. The network resources allocated and assigned in block **506** include, for example, one or more time periods and one or more frequency ranges of signals in communication infrastructure **110**. In one example of block **506**, (i) controller **112** identifies **604** network resources **606** of communication network **102** for use by termination device **108**(**1**) to perform a ranging and/or sounding process, (ii) controller **112** allocates **608** network resources **606** to termination device **108**(**1**), such as by updating a network resource allocation record (not shown) of network hub **106**, and (iii) controller **112** sends a message **610** to termination device **108**(**1**) including assignment of network resources **606** to termination device **108**(**1**), as illustrated in FIG. **6**.

[0051] In a block **508** of method **500**, the selected termination device **108** sends one or more calibrated ranging and/or sounding signals into communication infrastructure **110** (i) during one or more times periods allocated in the assigned network resources and (ii) within one or more frequency ranges allocated in the assigned network resources. Additionally, the selected termination device **108** listens for signals, i.e., it monitors communication infrastructure **110** for presence of signals, that are within at least the one or more frequency ranges encompassing the ranging and/or sounding signals, at least partially during the one or more time periods allocated in the assigned network resources. In one example of block **508**, termination device **108**(**1**) sends a calibrated ranging or sounding signal **612** into communication infrastructure **110**, as illustrated in FIG. **6**. A portion of ranging or sounding signal **612** returns to termination device **108**(**1**) via communication infrastructure **110** as a reflection signal **614** in response to ranging or sounding signal **612** encountering discontinuity **302**, as also illustrated in FIG. **6**. A portion **616** of ranging or sounding signal **612** remaining after generation of reflection signal **614** at discontinuity **302** continues to network hub **106** (and other elements of communication network **102** not shown in FIG. **6**) via communication infrastructure **110**.

[0052] In a block **510** of method **500**, the selected termination device **108** monitors communication infrastructure **110** and captures any signals received by the selected termination device **108** during, and optionally after, the one or more times periods allocated in the assigned network resources. The signals captured by the selected termination device **108** in block **510** include, for example, (i) one or more reflection signals resulting from the ranging and/or sounding signals sent in block **508** encountering one or more features, such as impairments, in communication infrastructure **110** and/or (ii) ingress of signals into communication infrastructure **110**, such as due to defects in communication infrastructure **110**. The selected termination device **108** monitors communication infrastructure **110** for signals at least within the one or more frequency ranges allocated in the assigned network resources, in block **510**. Additionally, in some embodiments, the selected termination device **108** further monitors communication infrastructure **110** for signals outside of the one or more frequency ranges allocated in the assigned network resources, such as to increase likelihood of detection of a non-linear feature and/or to enable detection of ingress of signals that are outside of the ranging or sounding signal frequency ranges.

[0053] In embodiments where communication network **102** is a cable communication network, the signals captured in block **510** include, for example, radio frequency reflection signals resulting from a network feature and/or undesired ingress of radio frequency signals into the cable communication network. The signals captured in block **510** may be expressed by the selected termination device **108** in a variety of manners. For example, the captured signals may be expressed as spectrum data (e.g., magnitude as a function of frequency), complex data (e.g., I, Q values), and/or RxMER per subcarrier. Additionally, the selected termination device **108** optionally converts any signals captured in block **510** into a form that facilitates processing of the captured signals. In one example of block **510**, termination device **108(1)** monitors **618** communication infrastructure **110** during one or more time periods allocated in assigned network resources **606**, and termination device **108(1)** captures (i) reflection signal **614** and (ii) signals **306** from ingress point **304**, while termination device **108(1)** monitors **618** communication infrastructure **110**, as illustrated in FIG. **6**.

[0054] In a block **512** of method **500**, the signals captured in block **510** are analyzed to identify one or more features, including but not limited to impairments, of communication infrastructure **110**. The analysis of block **512** is performed, for example, by the selected termination device **108**, controller **112**, and/or one or more other elements that are within, or external to, communication network **102**. In one example of block **512**, termination device **108(1)** and/or controller **112** analyze reflection signal **614** using a method similar to one or more of the analysis methods discussed above with respect to block **212** of method **200** but with (i) transmitted signal **414** replaced with ranging or sounding signal **612** and (ii) reflection signal **416** replaced with reflection signal **614**. As another example of block **512**, termination device **108(1)** and/or controller **112** analyze signals **306** to determine ingress using a method similar to that discussed above with respect to block **212** of method **200**.

[0055] FIG. **7** is a flow chart of method **700** for identifying a feature, such as an impairment, in a communication network, which is performed by certain embodiments of communication network **102**. In a block **702** of method **700**, a termination device **108** receives a message specifying an assignment of first network resources of the communication network to the first termination device. In one example of block **702**, termination device **108(1)** receives message **412** including assignment of network resources **408**, as illustrated in FIG. **4**. As another example of block **702**, termination device **108(1)** receives message **610** including assignment of network resources **606** to termination device **108(1)**, as illustrated in FIG. **6**.

[0056] In a block **704** of method **700**, the termination device **108** of block **702** sends one or more transmitted signals into infrastructure of the communication network at one or more times specified by the assignment of the first network resources, where the one or more transmitted signals are within one or more frequency ranges specified by the assignment of the first network resources. The one or more transmitted signals include, for example, a calibrated pilot signal, a ranging signal, a sounding signal, and/or communication traffic bearing signals. In one example of block **704**, termination device **108**(**1**) sends transmitted signal **414** into communication infrastructure **110**, as illustrated in FIG. **4**. As another example, of block **704**, termination device **108**(**1**) sends ranging or sounding signal **612** into communication infrastructure **110**, as illustrated in FIG. **6**.

[0057] In a block **706** of method **700**, the termination device **108** of blocks **702** and **704** captures one or more reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by one or more features, such as impairments, of the communication network. In one example of block **706**, termination device **108**(**1**) captures reflection signal **416**, as illustrated in FIG. **4**. In another example of block **706**, termination device **108**(**1**) captures reflection signal **614**, as illustrated in FIG. **6**. The signals captured in block **706** may be expressed by the termination device **108** in a variety of manners. For example, the captured signals may be expressed as spectrum data (e.g., magnitude as a function of frequency), complex data (e.g., I, Q values), and/or RxMER per subcarrier. Additionally, the termination device **108** optionally converts any signals captured in block **706** into a form that facilitates processing of the captured signals.

[0058] Method **700** optionally proceeds from block **706** to an additional block (not shown) where the termination device that captured the one or more reflection signals in block **706**, and/or controller **112**, analyze the captured reflection signals, such as using TDR or another feature identification technique. For example, in certain embodiments, the termination device **108** and/or controller **112** perform one or more the analysis methods discussed above with respect to block **212** of method **200** using the one or more reflection signals captured in block **706**.

[0059] Referring again to FIG. **1**, certain embodiments of communication network **102** are configured such that at least two termination devices **108** monitor communication infrastructure **110** for presence of signals, such as signals reflected by an impairment or other feature in communication infrastructure **110** or ingress signals, to identify an impairment or other feature in communication network **102**. For example, FIG. **8** is a flow chart of a method **800** for identifying a feature, such as an impairment, in a communication network, which is performed by certain embodiments of communication network **102**. Method **800** is discussed below with respect to the example impairments of FIG. **3** as well as with respect to a signal flow diagram **900** of FIG. **9** which illustrates examples of signal flows while executing method **800**. Signal flow diagram **900** includes vertical lines logically representing each of network hub **106** (including controller **112**), termination device **108**(**2**), termination device **108**(**3**), termination device **108**(**4**), and discontinuity **302**. Other elements of communication network **102** are not shown in FIG. **9** for illustrative clarity, and FIG. **9** should not be considered as being to scale. An arrow **901** in FIG. **9** represents increasing time in the direction of the arrow.

[0060] In a block **802** of method **800**, one or more termination device **108** are selected to serve as receivers over a predetermined time duration and over a predetermined frequency range, to perform a feature search. Additionally, one or more transmitted signal sources are selected in block **802** to provide one or more transmitted signals (e.g., a calibrated pilot signal, a ranging signal, a sounding signal, and/or communication traffic bearing signals) for the feature search. The transmitted signal sources are, for example, one or more of the termination devices **108** selected to serve as receivers, one or more termination devices **108** not selected to serve as receivers, network hub **106**, and/or an optional signal generator (not shown), such as a stand-alone signal generator connected to communication infrastructure **110**. In some embodiments, controller **112** causes termination devices **108** of communication network **102** to perform a ranging process and/or a sounding process to identify groups of termination devices **108** that are within communication range of each other via communication infrastructure **110**. Identification of groups of termination devices that are within communication range of each other may then be used to select devices in block **802** to ensure that all selected devices are in communication range of each other. For example, controller **112** may select in block **802** solely devices (e.g., termination devices **108**) from a single group of devices that are within communication range of each other, as determined by a ranging process and/or a sounding process. In one example of block **802**, termination devices **108**(**2**) and **108**(**3**) are selected to serve as receivers, and termination device **108**(**4**) is selected to serve as a transmitted signal source.

[0061] In a block **804** of method **800**, controller **112** notifies the devices selected in block **802**, henceforth referred to as "selected devices" in the method **800** discussion, that they will participate in a common feature search. In one example of block **804**, (i) controller **112** sends a message **902** to termination device **108**(**2**) notifying termination device **108**(**2**) that it will participate in the feature search along with termination devices **108**(**3**) and **108**(**4**), (ii) controller **112** sends a message **904** to termination device **108**(**3**) notifying termination device **108**(**3**) that it will participate in the feature search along with termination devices **108**(**2**) and **108**(**4**), and (iii) controller **112** sends a message **906** to termination device **108**(**4**) notifying termination device **108**(**4**) that it will participate in the feature search along with termination devices **108**(**3**) and **108**(**2**), as illustrated in FIG. **9**. In another example of block **804**, controller **112** sends a common message (not shown) to each of termination devices **108**(**2**), **108**(**3**), and **108**(**4**) notifying these three termination devices that they will participate in the feature search.

[0062] In a block **806** of method **800**, (i) controller **112** identifies network resources to assign to the selected devices, (ii) controller **112** allocates the identified network resources to the selected devices, (iii) controller **112** informs the selected devices that they have been assigned the identified network resources, (iv) controller **112** instructs the selected termination devices how to use the assigned network resources in the feature search, and (v) controller **112** optionally informs other devices (e.g., non-selected termination devices **108**) of the upcoming feature search. The network resources allocated and assigned in block **806** include, for example, one or more time periods and one or more frequency ranges of signals in communication infrastructure **110**. In embodiments where communication network **102** is a cable communication network (e.g., including coaxial electrical cables or a hybrid of optical cables and coaxial electrical cables), the network resources assigned in block **806** include, for example, one or more idle slots of the cable communication network. Controller **112** instructs the selected termination devices how to use the assigned network resources, by example, by (i) instructing the one or more devices selected to serve as receivers to monitor communication infrastructure **110** for signals over one or more time periods, and within one or more frequency ranges, specified in the assigned network resources, and (ii) instructing the one or more devices selected to serve as transmitted signal sources to send one or more transmitted signals into communication infrastructure **110** over the one or more time periods, and within the one or more frequency ranges, specified in the assigned network resources.

[0063] In one example of block **806**, (i) controller **112** identifies **908** network resources (NR) **910** of communication network **102** to allocate for the feature search, (ii) controller **112** allocates **912** network resources **910** to the selected devices, such as by updating a network resource allocation record (not shown) of network hub **106**, (iii) controller **112** sends messages **914**, **916**, and **918** to termination devices **108**(**2**), **108**(**3**), and **108**(**4**), respectively, including assignment of network resources **910** to termination devices **108**(**2**), **108**(**3**), and **108**(**4**), as illustrated in FIG. **9**. Messages **914** and **916** also instruct termination devices **108**(**2**) and **108**(**3**), respectively, to monitor communication infrastructure **110** for signals over one or more time periods, and within one or more frequency ranges, specified in the network resources **910**. In some alternate embodiments, messages **914** and **916** additionally instruct termination devices **108**(**2**) and **108**(**3**), respectively, to monitor communication infrastructure **110** for signals outside of the one or more frequency ranges specified in the network resources **910**, such as to enable identification of a non-linear feature, e.g., a non-linear impairment, and/or ingress of signals into communication infrastructure **110**. Additionally, message **918** instructs termination device **108**(**4**) to send one or more transmitted signals into communication infrastructure **110** over the one or more time periods, and within the one or more frequency ranges, specified in network resources **910**.

[0064] In a block **808** of method **800**, the one or more devices selected to serve as transmitted signal sources send one or more transmitted signals (e.g., pilot signals, ranging signals, sounding signals, and/or communication traffic bearing signals) into communication infrastructure **110** (i) during one or more times periods allocated in the assigned network resources and (ii) within one or more frequency ranges allocated in the assigned network resources. Additionally, the one or more termination devices **108** selected to serve as receivers listen for signals, i.e., they monitor communication infrastructure **110** for presence of signals, that are within at least the one or more frequency ranges encompassing the transmitted signals, at least partially during the one or more time periods allocated in the assigned network resources. In a manner similar to that discussed above with respect to block **208** of method **200**, the one or more transmitted signals may be short duration burst signals. Additionally, the devices selected to serve as transmitted signal sources may send transmitted signals into communication infrastructure **110** over multiple frequencies, such as according to a pattern of increasing frequencies where transmitted signal frequency increases with increasing time, a pattern of decreasing frequencies where transmitted signal frequency decreases with increasing time, a random pattern where transmitted signal frequency randomly varies with time, or any other pattern where transmitted signal frequency is a function of time. The devices selected to serve as transmitted signal sources are optionally configured to send a plurality of transmitted signals into communication infrastructure **110** over the time period allocated in the assigned network resources, for example, to help increase accuracy of the feature search, such as by enabling confirmation of reflection signals resulting from a feature or by enabling comparison of multiple reflection signals resulting from a feature.

[0065] In one example of block **808**, termination device **108**(**4**) sends a transmitted signal **920** into communication infrastructure **110**, as illustrated in FIG. **9**. Transmitted signal **920** reaches network hub **106**, termination devices **108**(**3**), and termination device **108**(**2**), as illustrated in FIG. **9**. However, transmitted signal **920** encounters discontinuity **302** before reaching termination device **108**(**1**) (not shown in FIG. **9**), resulting in generation of reflection signal **922**, as illustrated in FIG. **9**. Network hub **106**, termination device **108**(**2**), termination device **108**(**3**), and termination device **108**(**4**) subsequently receive reflection signal **922** via communication infrastructure **110**.

[0066] In a block **810** of method **800**, the termination devices **108** selected to serve as receivers monitor communication infrastructure **110** and capture any signals received by the termination devices **108** during, and optionally after, the one or more times periods allocated in the assigned network resources. The signals captured by the termination devices **108** selected to serve as receivers in block **810** include, for example, (i) one or more reflection signals resulting from the one or more transmitted signals sent in block **808** encountering one or more features, such as impairments, in communication infrastructure **110** and/or (ii) ingress of signals into communication infrastructure **110**, such as due to defects in communication infrastructure **110**. The termination devices **108** selected to serve as receivers monitor communication infrastructure **110** for signals at least within the one or more frequency ranges allocated in the assigned network resources, in block **810**. Additionally, in some embodiments, the termination devices **108** selected to serve as receivers further monitor communication infrastructure **110** for signals outside of the one or more frequency ranges allocated in the assigned network resources, such as to increase likelihood of detection of a non-linear feature and/or to enable detection of ingress of signals that are outside of the transmitted signal frequency ranges. In embodiments where communication network **102** is a cable communication network, the signals captured in block **810** include, for example, radio frequency reflection signals resulting from a network feature and/or undesired ingress of radio frequency signals into the cable communication network. In one example of block **810**, (i) termination device **108**(**2**) monitors **924** communication infrastructure **110** during one or more times periods allocated in assigned network resources **910** and captures reflection signal **922** and (ii) termination device **108**(**3**) monitors **926** communication infrastructure **110** during one or more times periods allocated in assigned network resources **910** and captures reflection signal **922**, as illustrated in FIG. **9**. It should be noted that characteristics of captured reflection signal **922** may vary between termination devices **108**(**2**) and **108**(**3**) due to their different respective locations among communication infrastructure **110**.

[0067] The signals captured in block **810** may be expressed by termination devices **108** in a variety of manners. For example, the captured signals may be expressed as spectrum data (e.g., magnitude as a function of frequency), complex data (e.g., I, Q values), and/or RxMER per subcarrier. Additionally, the one or more termination devices **108** capturing signals in block **810** optionally convert the signals into a form that facilitates processing of the captured signals.

[0068] In a block **812** of method **800**, the signals captured in block **810** are analyzed to identify one or more features, such as impairments, of communication infrastructure **110**. The analysis of block **810** is performed, for example, by one or more selected termination device **108**, controller **112**, and/or one or more other elements that are within, or external to, communication network **102**. For example, in certain embodiments, one or more termination devices **108** and/or controller **112** analyze a reflection signal captured in block **810** using a method similar to one or more of the analysis methods discussed above with respect to block **212** of method **200**, but with reflection signal **416** replaced with a reflection signal captured in block **810**. As another example, in certain embodiments, one or more termination devices **108** and/or controller **112** may determine that a signal captured in block **810** corresponds to an ingress signal using a procedure similar to that discussed above with respect to block **212** of FIG. **2** by (i) substituting transmitted signal **414** with a transmitted signal sent in block **808** and (ii) and signals **306** with signals captured in block **810**.

[0069] It should be noted that signals may be captured by two or more terminations devices **108** in block **810**. Respective signals captured by different termination devices **108** may be separately analyzed or they may be jointly analyzed. Respective signals captured by two or more termination devices in block **810** may correspond to a common feature, such as a common discontinuity or a common ingress point, such as in the example of FIG. **9** where reflection signals **922** captured by termination devices **108**(**2**) and **108**(**3**) correspond to common discontinuity **302**. Alternately, respective signals captured by two or more termination devices in block **810** may correspond to different respective network features. Accordingly, in particular embodiments, controller **112** and/or one or more termination devices **108** are configured to compare, such as using a pattern matching procedure, respective signals captured by two or more termination devices **108** at a common time, or captured by two or more termination devices **108** within a particular time frame, to determine one or more similarity values representing similarity of each of the captured signals to each of the other captured signals. In these embodiments, controller **112** and/or one or more termination devices **108** are configured to (i) determine that the captured signals correspond to a common feature in communication infrastructure **110** in response to the one or more similarity values being at least a predetermined threshold value and (ii) determine that the captured signals correspond to different respective features in communication infrastructure **110** in response to the one or more similarity values being less than the predetermined threshold value.

[0070] Determination of whether respective signals captured by two or more termination devices **108** in block **810** correspond to a common feature may facilitate locating the feature. For example, if respective signals captured by two or more termination devices **108** correspond to a common feature, the feature would necessarily be in a portion of communication infrastructure **110** serving each of the termination devices **108** capturing signals in block **810**. Conversely, if respective signals captured by two or more termination devices **108** in block **810** do not correspond to different respective features, portions of communication infrastructure **110** serving all of the termination devices **108** could potentially be ruled out as possible locations of the features. Additionally, determination of whether respective signals captured by two or more termination devices **108** correspond to a common feature may be used to determine likelihood of the termination devices **108** being close to each other in communication infrastructure **110**. Specifically, if the respective signals captured by two or more termination devices **108** in block **810** correspond to a common feature, it may be concluded that the termination devices **108** are likely relatively close to each other in communication infrastructure **110**. On the other hand, if the respective signals captured by two or more termination devices **108** in block **810** do not correspond to a common feature, it may be concluded that the termination devices **108** are not necessarily close to each other in communication infrastructure **110**.

[0071] In cases where respective signals captured by two or more termination devices **108** in block **810** correspond to a common feature, such as a common discontinuity, controller **112** and/or one or more termination devices **108** may determine a respective distance from each termination device **108** to the common feature using a TDR procedure. For instance, controller **112** and/or one or more termination devices **108** may determine a respective distance from each termination device **108**(**2**) and **108**(**3**) to discontinuity **302** in the FIG. **9** example by performing a TDR procedure. In particular, controller **112** and/or one or more termination devices **108** may determine distance of discontinuity **302** from termination device **108**(**2**) by dividing (i) a difference between a time when termination device **108**(**2**) receives reflection signal **922** and a time when termination device **108**(**2**) receives transmitted signal **920** by (ii) propagation velocity of signals along communication infrastructure **110**. Similarly, controller **112** and/or one or more termination devices **108** may determine distance of discontinuity **302** from termination device **108**(**3**) by dividing (i) a difference between a time when termination device **108**(**3**) receives reflection signal **922** and a time when termination device **108**(**3**) receives transmitted signal **920** by (ii) propagation velocity of signals along communication infrastructure **110**. Controller **112** and/or one or more termination devices **108** may estimate a location of a feature in communication infrastructure **110** via triangulation using respective distances of multiple termination devices **108** from the feature.

[0072] Additionally, in cases where respective signals captured by two or more termination devices **108** in block **810** correspond to a common feature in the form of a common ingress point, controller **112** and/or one or more termination devices **108** may determine a relative distance from each termination device **108** to the common ingress point from difference between respective times that each termination device **108** captures ingress signals. For example, controller **112** and/or one or more termination devices **108** may determine that a first termination device **108** is closer to the common ingress point than a second termination device **108** in response to the first termination device **108** capturing ingress signals from the common ingress point before the second termination device **108** captures ingress signals from the common ingress point. As another example, controller **112** and/or one or more termination devices **108** may determine that a first termination device **108** is further from the common ingress point than a second termination device **108** in response to the first termination device **108** capturing ingress signals from the common ingress point after the second termination device **108** captures ingress signals from the common ingress point.

[0073] Referring again to FIG. **1**, certain embodiments of communication network **102** are configured such that at least one termination device **108** is capable of effectively reflecting signals that it captures from communication infrastructure **110** back into communication infrastructure **110** as loopback signals, to further support identification of a feature in communication network **102**. For example, FIG. **10** is a flow chart of a method **1000** for identifying a feature in a communication network, which is performed by certain embodiments of communication network **102** including a termination device **108** that is configured to effectively reflect a received signal. Method **1000** is discussed below with respect to the example impairments of FIG. **3**, although it is understood that method **1000** could be used to identify other impairments. Method **1000** is additionally discussed with respect to a signal flow diagram **1100** of FIG. **11**, which illustrates examples of signal flows while executing method **1000**. Signal flow diagram **1100** includes vertical lines logically representing each of network hub **106** (including controller **112**), termination device **108**(**1**), termination device **108**(**2**), and discontinuity **302**. Other elements of communication network **102** are not shown in FIG. **11** for illustrative clarity, and FIG. **11** should not be considered as being to scale. An arrow **1101** in FIG. **11** represents increasing time in the direction of the arrow.

[0074] In a block **1002** of method **1000**, a first termination device **108** receives a message instructing the first termination device to monitor communication network infrastructure for presence of signals within a specified time range and within a specified frequency range. As one example of block **1002**, termination device **108**(**1**) receives a message **1102** from controller **112** instructing termination device **108**(**1**) to monitor communication infrastructure **110** for presence of signals within a specified time range and within a specified frequency range, as illustrated in FIG. **11**.

[0075] In a block **1004** the first termination device captures a first transmitted signal, such as a pilot signal, a ranging signal, a sounding signal, or a communication traffic bearing signal, during the time frame specified in the message of block **1002**. In one example of block **1004**, termination device **108**(**1**) monitors **1104** communication infrastructure **110** during the time range specified in message **1102** and captures a transmitted signal **1106** while monitoring **1104** communication infrastructure **110**, as illustrated in FIG. **11**. While outside the scope of block **1004**, FIG. **11** illustrates termination device **108**(**2**) sending a transmitted signal **1108** into communication infrastructure **110** where (i) a portion of transmitted signal **1108** is reflected back to termination device **108**(**2**), and other elements of communication network **102**, as a reflection signal **1110**, and (ii) a remaining portion of transmitted signal **1108** continues to termination device **108**(**1**) as transmitted signal **1106**. It is understood, though, that a device other than termination device **108**(**2**) could send a transmitted signal into communication infrastructure **110**.

[0076] In a block **1006** of method **1000**, the first termination device **108** sends a first loopback signal into communication infrastructure **110** in response to capturing the first transmitted signal in block **1004**. The first loopback signal is a function of the first transmitted signal captured in block **1004**. For example, the first loopback signal may have the same frequency and the same amplitude as the transmitted signal captured in block **1004**. As another example, the first loopback signal may be different from, but a function of, the transmitted signal captured in block **1004**, such as to encode information in the first loopback signal. For example, in some embodiments, amplitude of the first loopback signal is a function of one or more (i) amplitude of the transmitted signal captured in block **1004** and (ii) phase of the transmitted signal captured in block **1004**, to encode the first loopback signal with information related to the amplitude and/or phase of the transmitted signal captured in block **1004**. As such, the first termination device **108** effectively reflects the first transmitted signal back into communication infrastructure **110**, although the reflection signal is not necessarily identical to the transmitted signal captured in block **1004**. The first termination device **108** performs the reflection, for example, in a physical layer of the first termination device **108**. In some other embodiments, the first termination device **108** performs the reflection at a higher level, such as by (i) converting the captured first transmitted signal from analog form to digital form, (ii) generating a digital loopback signal that is a function of the digitized captured first transmitted signal, such as using software and/or firmware, and (iii) converting the digital loopback signal to an analog loopback signal for sending into communication infrastructure **110**. In certain embodiments, the first termination device **108** is capable of generating a loopback signal using more than one method, and in these embodiments, message **1102** may instruct the first termination device **108** which method to use when generating the loopback signal. Alternately, the first termination device **108** may generate multiple instances of the loopback signal using different respective methods.

[0077] In certain embodiments, the first termination device **108** sends the first loopback signal into communication infrastructure **110** essentially immediately after capturing the first transmitted signal in block **1004**, while in some other embodiments, first termination device **108** sends the first loopback signal into communication infrastructure **110** after (i) capturing the first transmitted signal in block **1004** and (ii) expiration of a delay period. The delay period, for example, is a function of internal processing of the first termination device **108**. In certain embodiments, the delay period is predetermined.

[0078] In one example of block **1006**, termination device **108(1)** sends a first loopback signal **1112** that is a function of captured transmitted signal **1106** into communication infrastructure **110**, as illustrated in FIG. **11**. First loopback signal **1112** encounters discontinuity **302**, and a portion of first loopback signal **1112** is reflected back to termination device **108(1)** as a first reflection signal **1114**, as illustrated in FIG. **11**. A remainder of first loopback signal **1112** continues to other elements of communication network **102** as first loopback signal **1116** after encountering discontinuity **302**, as further illustrated in FIG. **11**.

[0079] In a block **1008** of method **1000**, the first termination device captures a first reflection signal from communication network infrastructure resulting from reflection of the first loopback signal by one or more features in the communication network infrastructure. In one example of block **1008**, termination device **108(1)** monitors **1118** communication infrastructure **110** for a predetermined amount of time after sending first loopback signal **1112** into communication infrastructure **110**, and termination device **108(1)** captures first reflection signal **1114** while monitoring **1118** communication infrastructure **110**, as illustrated in FIG. **11**.

[0080] Method **1000** optionally further includes blocks **1010** and **1012**. In block **1010**, the first termination device **108** sends a second loopback signal into communication infrastructure **110** in response to capturing the first reflection signal in block **1008**. The second loopback signal is a function of the first reflection signal captured in block **1008**. For example, the second loopback signal may have the same frequency and the same amplitude as the first reflection signal captured in block **1008**, or amplitude of the second loopback signal may be a function of amplitude and/or of phase of the first reflection signal captured in block **1008**. As such, the first termination device **108** effectively reflects the first reflection signal back into communication infrastructure **110**, although the reflected signal is not necessarily identical to the first reflection signal captured in block **1008**. In one example of block **1010**, termination device **108(1)** sends a second loopback signal **1120** that is a function of captured first reflection signal **1114** into communication infrastructure **110**, as illustrated in FIG. **11**. Second loopback signal **1120** encounters discontinuity **302**, and a portion of second loopback signal **1120** is reflected back to termination device **108(1)** as a second reflection signal **1122**, as illustrated in FIG. **11**. A remainder of second loopback signal **1120** continues to other elements of communication network **102** as second loopback signal **1124** after encountering discontinuity **302**, as further illustrated in FIG. **11**.

[0081] In a block **1012** of method **1000**, the first termination device captures a second reflection signal from communication network infrastructure resulting from reflection of the second loopback signal by one or more features in the communication network infrastructure. In one example of block **1012**, termination device **108(1)** monitors **1126** communication infrastructure **110** for a predetermined amount of time after sending second loopback signal **1120** into communication infrastructure **110**, and termination device **108(1)** captures second reflection signal **1122** while monitoring **1126** communication infrastructure **110**, as illustrated in FIG. **11**. Method **1000** could be modified so that reflections resulting from loopback signals encountering a feature are captured by one or more termination devices **108** other than, or in addition to, the termination device sending loopback signals into communication infrastructure **110**.

[0082] Signals captured in method **1000** can be used to determine respective distances from one or more termination devices **108** to the feature in communication infrastructure **110**. For example, referring again to FIG. **11**, termination device **108(2)**, controller **112**, and/or one or more other elements of communication network **102** may determine a distance X between termination device **108(2)** and discontinuity **302** by dividing (i) a difference between a time when termination device **108(2)** captures reflection signal **1110** and a time when termination device **108(2)** sends transmitted signal **1108** into communication infrastructure **110** by (ii) propagation velocity of signals in communication infrastructure **110**.

[0083] As another example, termination device **108**(**1**), controller **112**, and/or one or more other elements of communication network **102** may determine a distance Y between termination device **108**(**1**) and discontinuity **302** by dividing (i) a difference between a time when termination device **108**(**1**) captures first reflection signal **1114** and a time when termination device **108**(**1**) sends first loopback signal **1112** into communication infrastructure **110** by (ii) propagation velocity of signals in communication infrastructure **110**. Furthermore, one or more termination devices **108**, controller **112**, and/or one or more other elements of communication network **102** may determine distance Y based on distance X and Z by subtracting distance X from distance Z. Similarly, one or more termination devices **108**, controller **112**, and/or one or more other elements of communication network **102** may determine distance X based on distance Y and Z by subtracting distance Y from distance Z. Distance Z may be known, or distance Z may be determined by communication network **102**. For example, in some embodiments, communication network **102** determines distance Z by (i) sending a transmitted signal between termination devices **108**(**1**) and **108**(**2**), such as a pilot signal, a ranging signal, or a sounding signal, (ii) determining a travel time for the transmitted signal to travel between termination devices **108**(**1**) and **108**(**2**), and (iii) dividing the travel time by a propagation velocity of signals in communication infrastructure **110**.

[0084] The signals captured in method **1000** can also be analyzed to determine one or more characteristics of the feature other than relative location of the feature. For example, referring again to FIG. **11**, termination device **108(1**), controller **112**, and/or one or more other elements of communication network **102** may determine reflection strength of discontinuity **302** by comparing amplitude of first reflection signal **1114** to amplitude of first loopback signal **1112**, where strength of discontinuity **302** is inversely proportional to the difference between the amplitude of first reflection signal **1114** and the amplitude of first loopback signal **1112**. As another example, termination device **108**(**2**), controller **112**, and/or one or more other elements of communication network **102** may determine strength of discontinuity **302** by comparing amplitude of reflection signal **1110** to amplitude of transmitted signal **1108**, where strength of discontinuity **302** is inversely proportional to the difference between the amplitude of reflection signal **1110** and the amplitude of transmitted signal **1108**.

[0085] Method **1000** could be modified to omit blocks **1010** and **1012**, and the first termination device **108** would therefore send only one loopback signal into communication infrastructure **110**. Alternately, method **1000** could be modified to include one or more additional sets of blocks analogous to optional blocks **1010** and **1012** so that method **1000**, and the first termination device **108** would therefore send three or more loopback signals into communication infrastructure **110**, depending on the quantity of additional sets of blocks included in method **1000**.

[0086] It should be noted that the effective reflection of signals received by the first termination device **108** in method **1000** results in an echo tunnel in communication infrastructure **110** between the first termination device **108** and the feature in communication infrastructure **110**. This feature of method **1000** can be leveraged to enhance identification of an impairment or other feature in communication infrastructure **110**. For example, consider again FIG. **11**. Termination device **108**(**2**), controller **112**, and/or one or more other elements of communication network **102** may determine that discontinuity **302** is located in communication infrastructure **110** between termination device **108**(**1**) and termination device **108**(**2**) in response to (i) termination device **108**(**2**) capturing reflection signal **1110** after sending transmitted signal **1108** and (ii) terminating device **108**(**1**) capturing reflection signal **1114** after sending loopback signal **1112** into communication infrastructure **110**. As another example, in certain embodiments, one or more termination devices **108**, controller **112**, and/or one or more other elements of communication network **102** are configured to determine distances X, Y, and Z of FIG. **11**, such as using the procedures discussed above. In these embodiments, one or more termination devices **108**, controller **112**, and/or one or more other elements of communication network **102**, are configured to determine that discontinuity **302** is located in communication infrastructure **110** between termination device **108**(**1**) and termination device **108**(**2**) in response to the sum of distance X and distance Y being equal to distance Z, or the sum of distance X and distance Y being within a predetermined percentage (e.g., +/−1 percent, +/−5 percent, +/−10 percent, etc.) of distance Z.

[0087] Some embodiments of the methods discussed above may be performed outside of a quiet period in the communication network. Accordingly, in certain embodiments, one or more termination devices **108** are configured to determine what portion of signals captured by the termination device **108** are from other termination devices **108**, thereby enabling the termination device **108** to determine what portion of captured signals are from reflection of transmitted signals sent by the termination device **108**. For example, in certain embodiments, one or more termination devices **108** are configured to send a first transmitted signal into communication infrastructure **110** at full power and subsequently send multiple transmitted signals into communication infrastructure **110** while reducing the transmit power of each subsequent transmitted signal until transmit power of the transmitted signals is zero. In these embodiments, the termination device **108** may (i) determine power of signals from other termination devices **108** based on power of signals captured by the termination device **108** after the power of the transmitted signals is zero and/or (ii) determine reflection strength of a feature in communication infrastructure **110** from a difference between power of a signal captured after sending the first transmitted signal into communication infrastructure **110** and power of a signal captured after power of the transmitted signals is zero.

[0088] Referring again to FIG. **1**, as discussed above, in some embodiments, communication network **102** is configured as a cable communication network. For example, FIG. **12** is a schematic diagram of a communication environment **1200**, which is an embodiment of communication environment **100** where communication network **102** is embodied by a cable communication network **1202**. Particular embodiments of cable communication network **1202** are configured to execute one or more of methods **200**, **500**, **700**, **800**, or **1000**, discussed above, to identify a feature, such as an impairment in cable communication network **1202**. Network hub **106** is embodied by a network hub **1206**, communication infrastructure **110** is embodied by communication infrastructure **1210**, and termination devices **108** are embodied by cable modems **1208**, in cable communication network **1202**. Network Hub **1206** includes a CMTS **1214**, an R-PHY device **1216**, and optical cable **1218**. While FIG. **12** depicts controller **112** as being part of CMTS **1214**, the configuration of controller **112** could vary in communication environment **1200** without departing from the scope hereof. For example, controller **112** could alternately be part of R-PHY device **1216** instead of being part of CMTS **1214**, or controller **112** could be distributed between CMTS **1214** and R-PHY device **1216**. As another example, controller **112** could be (i) separate from both of CMTS **1214** and R-PHY device **1216** within network hub **1206**. As a further example, controller **112** could be external to network hub **1206**. While not required, it is anticipated that network hub **1206** will frequently include additional R-PHY devices and associated communication infrastructure.

[0089] Optical cable **1218** communicatively couples CMTS **1214** and R-PHY device **1216**. Although optical cable **1218** is depicted as having a short length, optical cable **1218** could have a long length, such as several kilometers, tens of kilometers, or more. As such, the elements of network hub **1206** could be distributed over a large physical area. CMTS **1214** interfaces network hub **1206** with external network resources **104**. Additionally, CMTS **1214** and R-PHY device **1216** are collectively configured to control communication between cable modems **1208** and network hub **1206** via communication infrastructure **1210**. CMTS **1214** is, for example, a hardware CMTS or a virtual CMTS implemented by one or more processors executing non-transitory instructions in the form of software and/or firmware stored in a data store. In some alternate embodiments, R-PHY device **1216** is omitted and the functions of R-PHY device **1216** are implemented by CMTS **1214**. In some other alternate embodiments, R-PHY device **1216** is replaced with a R-MAC-PHY device or one or more other devices.

[0090] Communication infrastructure **1210** includes a fiber node **1220**, a tap **1222**, a tap **1224**, a tap **1226**, optical cable **1228**, a hardline coaxial electrical cable **1230**, a hardline coaxial electrical cable **1232**, a hardline coaxial electrical cable **1234**, a drop cable **1236**, a drop cable **1238**, a drop cable **1240**, and a drop cable **1242**. Optical cable **1228** communicatively couples fiber node **1220** to R-PHY device **1216**, and hardline coaxial electrical cable **1230** communicatively couples tap **1222** to fiber node **1220**. Hardline coaxial electrical cable **1232** communicatively couples tap **1224** to tap **1222**, and hardline coaxial electrical cable **1234** communicatively couples tap **1226** to tap **1222**. Drop cable **1236** communicatively couples cable modem **1208**(**1**) to tap **1224**, and drop cable **1238** communicatively couples cable modem **1208**(**2**) to tap **1222**. Drop cable **1240** communicatively couples cable modem **1208**(**3**) to tap **1226**, and drop cable **1242** communicatively couples cable modem **1208**(**4**) to tap **1226**.
[0091] Fiber node **1220** translates communication signals in the optical domain on optical cable **1228** to communication signals in the electrical domain on hardline coaxial electrical cable **1230** and vice versa. In some alternate embodiments, fiber node **1220** and optical cable **1228** are omitted, and hardline coaxial electrical cable **1230** communicatively couples tap **1222** to R-PHY device **1216**. Additionally, communication infrastructure **1210** could include additional elements, such as amplifiers and/or power inserters. FIG. **12** depicts impairments **1244** and **1246** which are examples of impairments **302** and **304** of FIG. **3**, respectively. Impairment **1244** is, for example, a discontinuity in hardline coaxial electrical cable **1232**, such as resulting from water intrusion into the cable. Impairment **1246** is, for example, an opening in hardline coaxial electrical cable **1230** allowing undesired ingress of radio frequency signals into communication infrastructure **1210**.
[0092] In particular embodiments, cable modems **1208** are FDX cable modems or frequency division duplex (FDD) cable modems. Each cable modem **1208** need not have the same configuration. For example, cable modem **1208**(**1**) could be an FDD cable modem, and cable modem **1208**(**2**) could be a FDX cable modem. In some embodiments, cable communication network **1202** is configured to operate according to a DOCSIS 3.1 communication standard, a DOCSIS 4.0 communication standard, and/or one or more successor cable communication standards. In certain embodiments, one or more cable modems are configured to use one or more of full band capture, upstream data analysis (UDA,) a symbol capture procedure, a pre-equalization procedure, and/or a channel estimation procedure, to capture signals to identify a feature, such as when cable communication network **1202** executes one of methods **200**, **500**, **700**, **800**, or **1000**.
[0093] The performance of one or more of methods **200**, **500**, **700**, **800**, or **1000** by cable communication network **1202** may advantageously help identify potential interference between cable modems **1208**, especially in embodiments where some cable modems **1208** operate according to a high performance communication standard (e.g., DOCSIS **4.0**) while other cable modems **1208** operate according to a lower performance communication standard (e.g., DOCSIS 3.1, DOCSIS 3.0, or DOCSIS 2.0). In particular, a cable modem **1208** operating at a high performance communication standard may generate communication signals having a higher energy level than communication signals generated by a cable modem **1208** operating at a lower performance communication standard. Consequently, the cable modem **1208** operating at the high performance communication standard may cause interference with the cable modem **1208** operating at the lower performance communication standard. Methods **200**, **500**, **700**, **800**, or **1000** can be used to identify this interference by (i) causing a cable modem **1208** to send transmitted signals into communication infrastructure **1210** having characteristics of signals generated by a cable modem **1208** operating at a high performance communication standard and (ii) monitoring operation of other cable modems **1208** to determine whether the cable modems **1208** experience performance degradation, such as increased forward error correction or decreased up-time duration, due to sending the transmitted signals into communication infrastructure **1210**.
[0094] For example, assume that cable modems **1208**(**1**) and **1208**(**2**) currently operate according to a DOCSIS 3.0 communication standard and it is proposed to upgrade cable modem **1208(2**) to a

DOCSIS 4.0 communication standard. Potential for interference between cable modems **1208**(**2**) and **1208**(**1**) if cable modem **1208**(**2**) operates according to the DOCSIS 4.0 communication standard may be determined, for example, by executing method **200** in cable communication network **1202** where (i) cable modem **1208**(**2**) is selected to perform feature search in block **202** and (ii) cable modem **1208**(**2**) sends transmitted signals into communication infrastructure **1210** in block **208** where the transmitted signals have characteristics of DOCSIS 4.0 communication signals. Operation of cable modem **1208**(**1**) can be simultaneously monitored for impaired operation resulting from cable modem **1208**(**2**) sending the transmitted signals into communication infrastructure **1210**.

[0095] In some embodiments, one or more cable modems **1208** are further configured to estimate distance between the cable modem **1208** and a network feature, e.g., impairment **1244**, using phase information, such as phase information derived from one or more of a channel estimation process, a pre-equalization process, and a symbol capture process. For example, a cable modem **1208** may use one or more of a channel estimation process, a pre-equalization process, and a symbol capture process to determine a complex channel response of the cable modem **1208** in a frequency domain. The cable modem **1208** may then transform the channel response from the frequency domain to a time domain, such as using a Fourier transform process. The resulting channel response in the time domain indicates signals that are out of phase, or delayed in time, due to a reflection, e.g., such as from an impairment or other network feature. In these embodiments, the cable modem **1208** divides the time delay by a signal propagation velocity in cable communication network **1202**, e.g., by signal propagation velocity along a coaxial electrical cable, to estimate a distance between the cable modem **1208** and the network feature causing signal reflection.

[0096] For example, FIG. **13** is a flow chart of a method for determining respective locations of one or more features of a communication network relative to a cable modem of the communication network, which is one example of how certain embodiments of cable modems **1208** may determine relative location of a network feature using phase information. In a block **1302** of method **1300**, controller **112** controls a cable modem **1208** and network hub **1206** of cable communication network **1202** to determine a channel response of cable communication network **1202** between the cable modem **1208** and network hub **1206** using one or more of (i) a channel estimation process, (ii) a pre-equalization process, and (iii) a symbol capture process. As known in the art, a channel estimation process includes a cable modem receiving a signal and performing a complex equalization process to recover information from the signal. The cable modem can subsequently recover a channel response from the signal. A pre-equalization process, on the other hand, includes a cable modem sending a signal to a CMTS and the cable modem adjusting its frequency response in accordance with instructions from the CMTS, thereby pre-equalizing uplink signals from the cable modem to account for cable plant issues. The cable modem may subsequently determine a channel response from the pre-equalization parameters. A symbol capture process, in turn, includes a cable modem capturing a predetermined symbol sent by a CMTS, and the cable modem can determine a channel response from the captured symbol, e.g., as a function of distortion of the captured symbol. In one example of block **1302**, controller **112** controls CMTS **1214** and cable modem **1208**(**1**) to determine a channel response between cable modem **1208**(**1**) and network hub **1206** using one or more of (i) a channel estimation process, (ii) a pre-equalization process, and (iii) a symbol capture process.

[0097] In a block **1304** of method **1300**, the channel response determined in block **1302** is converted from a frequency domain to a time domain. In one example of block **1304**, cable modem **1208**(**1**) and/or controller **112** convert the channel response between cable modem **1208**(**1**) and network hub **1206** from a frequency domain to a time domain using a Fourier transform process. In a block **1306** of method **1300**, the phase shift information is determined from the channel response in the time domain from block **1304**. In one example of block **1306**, cable modem **1208**(**1**) and/or controller **112** determine phase shift information from the channel response between cable modem

1208(**1**) and network hub **1206** in the time domain. In a block **1308** of method **1300**, respective distances of the one or more features of the communication network from the cable modem **1208** are estimated at least partially based on the signal phase shift information. In one example of block **1308**, cable modem **1208**(**1**) and/or controller **112** (i) determine time delay resulting from impairment **1244** from the phase shift information determined in block **1306** and (ii) estimate the distance of impairment **1244** by dividing the time delay by signal propagation velocity along a coaxial electrical cable.

[0098] Referring again to FIG. **1**, as discussed above, in some embodiments, communication network **102** is configured as a PON communication network. For example, FIG. **14** is schematic diagram of a communication environment **1400**, which is an embodiment of communication environment **100** where communication network **102** is embodied by a PON communication network **1402**. Particular embodiments of PON communication network **1402** are configured to execute one or more of methods **200**, **500**, **700**, **800**, or **1000**, discussed above, to identify a feature, such as an impairment, in PON communication network **1402**. Network hub **106** is embodied by a network hub **1406**, communication infrastructure **110** is embodied by communication infrastructure **1410**, and termination devices **108** are embodied by ONTs/ONUs **1408**, in PON communication network **1402**. Network hub **1406** includes an OLT **1414**. While FIG. **14** depicts controller **112** as being part of OLT **1414**, the configuration of controller **112** could vary in communication environment **1400** without departing from the scope hereof. For example, controller **112** could alternately be separate from OLT **1414** within network hub **1406**. As another example, controller **112** could be external to network hub **1406**. OLT **1414** interfaces network hub **1406** with external network resources **104**, and OLT **1414** also controls communication between ONTs/ONUs **1408** and network hub **1406** via communication infrastructure **1410**.

[0099] Communication infrastructure **1410** includes a splitter **1416**, a splitter **1418**, optical cable **1420**, optical cable **1422**, optical cable **1424**, optical cable **1426**, optical cable **1428**, and optical cable **1430**. Optical cable **1420** communicatively couples splitter **1416** to OLT **1414**, and optical cable **1424** communicatively couples splitter **1418** to splitter **1416**. Optical cable **1422** communicatively couples ONT/ONU **1408**(**1**) to splitter **1416**, and optical cable **1426** communicatively couples ONT/ONU **1408**(**2**) to splitter **1416**. Optical cable **1428** communicatively couples ONT/ONU **1408**(**3**) to splitter **1418**, and optical cable **1430** communicatively couples ONT/ONU **1408**(**4**) to splitter **1418**.

[0100] Each ONT/ONU **1408** may be either an ONT or an ONU. In some embodiments, PON communication network **1402** operates according to one or more of an Ethernet passive optical network (EPON) protocol, a radio frequency of over glass (RFOG or RFoG) protocol, a Gigabit-capable passive optical network (GPON) protocol, an XG-PON protocol, an XGS-PON protocol, and successors of any of foregoing protocols. FIG. **14** depicts an impairment **1432** which is an example of impairment **302** of FIG. **3**. Impairment **1432** is, for example, a crack in optical cable **1422** causing a change in index of refraction of optical cable **1422** at the location of impairment **1432**.

[0101] Referring again to FIG. **1**, as discussed above, in some embodiments, communication network **102** is configured as a coherent optical communication network. For example, FIG. **15** is schematic diagram of a communication environment **1500**, which is an embodiment of communication environment **100** where communication network **102** is embodied by a coherent optical communication network **1502**. Particular embodiments of coherent optical communication network **1502** are configured to execute one or more of methods **200**, **500**, **700**, **800**, or **1000**, discussed above, to identify a feature, such as an impairment, in coherent optical communication network **1502**. Network hub **106** is embodied by a network hub **1506**, communication infrastructure **110** is embodied by communication infrastructure **1510**, and termination devices **108** are embodied by a coherent transceiver **1508**, in coherent optical communication network **1502**. In contrast to cable communication network **1202** of FIG. **12** and PON communication network **1402** of FIG. **14**, coherent optical communication network **1502** is a transmission communication network instead of an access communication network. Accordingly, coherent optical communication network **1502** includes only a single termination device, i.e., coherent transceiver **1508**. Coherent optical communication network **1502** is used, for example, to transmit communication signals over long distances or to transmit communication signals over short distances in data centers, such as in hyperscaler data centers.

[0102] Network hub **1506** includes a coherent transceiver **1514**, and communication infrastructure **1510** includes an optical cable **1516** communicatively coupling coherent transceiver **1514** and coherent transceiver **1508**. While FIG. **15** depicts controller **112** as being part of coherent transceiver **1514**, the configuration of controller **112** could vary in communication environment **1500** without departing from the scope hereof. For example, controller **112** could alternately be separate from coherent transceiver **1514** within network hub **1506**. As another example, controller **112** could be external to network hub **1506**. Coherent transceiver **1514** interfaces network hub **1506** with external network resources **104**. Additionally, coherent transceiver **1514** is configured to modulate optical signals for transmission to coherent transceiver **1508** via optical cable **1516** by modulating each of amplitude, phase, and polarization of the optical signals. Furthermore, coherent transceiver **1514** is configured to demodulate coherent optical signals received from coherent transceiver **1508** via optical cable **1516**. Coherent transceiver **1508** is configured to modulate optical signals for transmission to coherent transceiver **1514** via optical cable **1516** by modulating each of amplitude, phase, and polarization of the optical signals. Additionally, coherent transceiver **1508** is configured to demodulate coherent optical signals received from coherent transceiver **1514** via optical cable **1516**. While coherent optical communication network **1502** is configured to support two-way communication, coherent optical communication network **1502** could be modified to support only one-way communication.

[0103] FIG. **15** illustrates an impairment **1518**, which is one example of an impairment that may be identified by coherent optical communication network **1502** executing one or more of methods **200**, **500**, **700**, **800**, or **1000**. Impairment **1518** is, for example, a crack in optical cable **1516** causing a change in index of refraction of optical cable **1516** at the location of impairment **1518**.

[0104] Referring again to FIG. **1**, as discussed above, in some embodiments, communication network **102** is configured as a wireless communication network. For example, FIG. **16** is a schematic diagram of a communication environment **1600**, which is an embodiment of communication environment **100** where communication network **102** is embodied by a wireless communication network **1602**. Particular embodiments of wireless communication network **1602** are configured to execute one or more of methods **200**, **500**, **700**, **800**, or **1000**, discussed above, to identify a feature, such as an impairment, in wireless communication network **1602**. Network hub **106** is embodied by a network hub **1606**, communication infrastructure **110** is embodied by communication infrastructure **1610**, and termination devices **108** are embodied by wireless modems **1608**, in wireless communication network **1602**. Network hub **1606** includes a wireless core network **1614**. While FIG. **16** depicts controller **112** as being part of wireless core network **1614**, the configuration of controller **112** could vary in communication environment **1600** without departing from the scope hereof. For example, controller **112** could alternately be separate from wireless core network **1614** within network hub **1606**. As another example, controller **112** could be external to network hub **1606**. Wireless core network **1614** interfaces network hub **1606** with external network resources **104**, and wireless core network **1614** also controls communication between wireless modems **1608** and network hub **1606** via communication infrastructure **1610**.

[0105] Communication infrastructure **1610** includes a wireless access point **1616** and a communication link **1618** communicatively coupling wireless access point **1616** to wireless core network **1614**. Communication infrastructure **1610** may also be considered to include free space (not labeled) for transmission of wireless communication signals. Communication link **1618** may include multiple communication mediums and/or communication hardware. Each wireless modem

1608 is configured to wirelessly communicate with wireless access point **1616** via respective wireless communication signals **1620**.

[0106] In some embodiments, wireless communication network **1602** is configured as a cellular wireless communication network, e.g., operating according to a Third Generation Partnership Project (3GPP) standard, such as a Long Term Evolution (LTE) standard, a Fifth Generation (5G) standard, a Sixth Generation (6G) standard, and/or successors thereof. In embodiments where wireless communication network **1602** is a cellular wireless communication network, wireless core network **1614** includes, for example, an evolved packet core, a 5G core network, and/or a 6G core network. Additionally, in some embodiments where wireless communication network **1602** is a cellular wireless communication network, wireless access point **1616** is part of a radio access network (RAN). In some other embodiments, wireless communication network **1602** operates according to an Institute of Electrical and Electronics Engineers (IEEE) standard, such as a Wi-Fi standard or a HaLow standard. Additionally, in certain other embodiments, wireless communication network **1602** is a Bluetooth wireless communication network, a satellite wireless communication network (e.g., using very low earth orbit (VLEO) satellites, low earth orbit (LEO) satellites, medium earth orbit (MEO) satellites, or geostationary equatorial orbit (GEO) satellites), a LoRa wireless communication network, a Zigbee wireless communication network, or a Z-wave wireless communication network. However, wireless communication network **1602** could be another type of wireless communication network without departing from the scope hereof.

[0107] FIG. **16** illustrates an impairment **1622**, which is one example of an impairment that may be identified by wireless communication network **1602** executing one or more of one or more of methods **200**, **500**, **700**, **800**, or **1000**. Impairment **1622** is, for example, fading of wireless communication signals **1620**(**1**) caused by a temporary or permanent obstacle (not shown) between wireless modem **1608**(**1**) and wireless access point **1616**.

Combinations of Features

[0108] Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations. [0109] (A1) A method for identifying a feature in a communication network includes (1) receiving, at a first termination device of the communication network, a message specifying an assignment of first network resources of the communication network to the first termination device, (2) sending, from the first termination device, one or more transmitted signals into infrastructure of the communication network at one or more times specified by the assignment of the first network resources, the one or more transmitted signals being within one or more frequency ranges specified by the assignment of the first network resources, and (3) capturing, at the first termination device, one or more reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by one or more features in the infrastructure of the communication network. [0110] (A2) In the method denoted as (A1), the one or more transmitted signals may include one or more pilot signals. [0111] (A3) In either one of the methods denoted as (A1) and (A2), the one or more transmitted signals may include at least one of (i) a ranging signal and (ii) a sounding signal. [0112] (A4) In any one of the methods denoted as (A1) through (A3), the one or more times specified by the assignment of the first network resources may correspond to one or more quiet periods of the communication network. [0113] (A5) In any one of the methods denoted as (A1) through (A4), (1) the first termination device may be a cable modem, and (2) the step of capturing the one or more reflection signals from the infrastructure of the communication network may include using upstream data analysis (UDA) of the cable modem to capture the one or more reflection signals. [0114] (A6) In any one of the methods denoted as (A1) through (A4), (1) the first termination device may be a cable modem, and (2) the step of capturing the one or more reflection signals from the infrastructure of the communication network may include using a symbol capture procedure at the cable modem. [0115] (A7) In any one of the methods denoted as (A1) through (A6), the first network resources may include one or more time periods and one or more frequency ranges. [0116] (A8) Any one of the methods denoted as (A1) through (A7) may further include determining one or more characteristics of the one or more features in the infrastructure of the communication network at least partially based on the one or more reflection signals captured by the first termination device. [0117] (A9) Any one of the methods denoted as (A1) through (A8) may further include determining respective locations of the one or more features in the infrastructure of the communication network relative to the first termination device at least partially based on one or more differences in time between (i) respective times that the one or more reflection signals are captured by the first termination device and (ii) respective times that the one or more transmitted signals are sent into the infrastructure of the communication network from the first termination device. [0118] (A10) Any one of the methods denoted as (A1) through (A9) may further include determining respective reflection strengths of the one or more features in the infrastructure of the communication network at least partially based on the one or more reflection signals captured by the first termination device from the infrastructure of the communication network. [0119] (A11) In any one of the methods denoted as (A1) through (A10), the one or more features of the communication network may include one or more discontinuities in one or more communication cables of the communication network. [0120] (A12) Any one of the methods denoted as (A1) through (A11) may further include determining presence of a non-linear feature in response to the one or more reflection signals captured by the first termination device from the infrastructure of the communication network being within a frequency range that is different from a frequency range of the one or more transmitted signals. [0121] (A13) Any one of the methods denoted as (A1) through (A12) may further include determining presence of signal ingress in response to one or more additional signals captured by the first termination device from the infrastructure of the communication network being within a frequency range that is different from a frequency range of the one or more transmitted signals. [0122] (A14) Any one of the methods denoted as (A1) through (A13) may further include, at the first termination device, sending a request for the first network resources to a network hub. [0123] (A15) In the method denoted as (A14) the network hub may include one or more of a cable modem termination system (CMTS), a remote physical layer (R-PHY) device, an optical line terminal (OLT), an ONT, an ONU, a BNG, and a core wireless communication network. [0124] (A16) Any one of the methods denoted as (A1) through (A14) may further include receiving the message specifying the assignment of the first network resources of the communication network to the first termination device from a network hub. [0125] (A17) In the method denoted as (A16), the network hub may include one or more of a cable modem termination system (CMTS), a remote physical layer (R-PHY) device, an optical line terminal (OLT), an ONT, an ONU, a BNG, and a core wireless communication network. [0126] (A18) In any one of the methods denoted as (A1) through (A14), (1) the communication network may be a cable communication network, and (2) the first termination device may be a cable modem connected to a coaxial electrical cable of the communication network. [0127] (A19) In any one of the methods denoted as (A1) through (A14), (1) the communication network may be an optical communication network, and (2) the first termination device may be one of an optical network termination (ONT) and an optical network unit (ONU) connected to an optical cable of the communication network. [0128] (A20) In any one of the methods denoted as (A1) through (A14), the communication network may be a coherent optical communication network. [0129] (A21) In any one of the methods denoted as (A1) through (A14), (1) the communication network may be a wireless communication network, and (2) the first termination device may be a wireless modem. [0130] (A22) In any one of the methods denoted as (A1) through (A21), the first termination device may be capable of receiving communication signals having respective frequencies of the one or more transmitted signals. [0131] (B1) A method for identifying a feature in a communication network includes (1) receiving, at a first termination device of the communication network, a message instructing the first termination device to monitor infrastructure of the communication network for presence of signals within a specified time period and within a specified frequency range, (2) capturing, at the first termination device, a transmitted signal during the specified time period, (3) sending, from the first termination device, a loopback signal into the infrastructure of the communication network, the loopback signal being a function of the transmitted signal captured by the first termination device, and (4) capturing, at the first termination device, one or more first reflection signals from the infrastructure of the communication network resulting from reflection of the loopback signal by one or more features of the infrastructure of the communication network. [0132] (B2) The method denoted as (B1) may further include sending, from a second termination device of the communication network, the transmitted signal into the infrastructure of the communication network. [0133] (B3) In the method denoted as (B2), the transmitted signal may include at least one of (i) a pilot signal, (ii) a ranging signal, (iii) a sounding signal, and (iv) a communication traffic bearing signal. [0134] (B4) Any one of the methods denoted as (B1) through (B3) may further include determining respective locations of the one or more features in the infrastructure of the communication network relative to the first termination device at least partially based on one or more differences in time between (i) respective times that the one or more first reflection signals are captured by the first termination device and (ii) respective times that the loopback signal is sent into the infrastructure of the communication network from the first termination device. [0135] (B5) Any one of the methods denoted as (B1) through (B4) may further include determining respective locations of the one or more features in the infrastructure of the communication network relative to a second termination device of the communication network at least partially based on differences between (i) distance between the second termination device and the first termination device and (ii) respective distances of the one or more features in the infrastructure of the communication network from the first termination device. [0136] (B6) Any one of the methods denoted as (B1) through (B5) may further include determining one or more characteristics of the one or more features in the infrastructure of the communication network at least partially based on the one or more first reflection signals captured by the first termination device from the infrastructure of the communication network. [0137] (B7) In any one of the methods denoted as (B1) through (B6), (1) the communication network may be a cable communication network and (2) the first termination device may be a cable modem connected to a coaxial electrical cable of the communication network. [0138] (B8) In any one of the methods denoted as (B1) through (B6), (1) the communication network may be an optical communication network and (2) the first termination device may be one of an optical network termination (ONT) and an optical network unit (ONU) connected to an optical cable of the communication network. [0139] (B9) In any one of the methods denoted as (B1) through (B6), the communication network may be a coherent optical communication network. [0140] (B10) In any one of the methods denoted as (B1) through (B6), (1) the communication network may be a wireless communication network and (2) the first termination device may be a wireless modem. [0141] (C1) A method for identifying a feature in a communication network includes (1) receiving, at a first termination device of the communication network, a message instructing the first termination device to monitor infrastructure of the communication network for presence of signals within a specified time period and within a specified frequency range, (2) receiving, at a second termination device of the communication network, a message instructing the second termination device to monitor the infrastructure of the communication network for presence of signals within the specified time period and within the specified frequency range, (3) capturing, at the first termination device, one or more first reflection signals from the infrastructure of the communication network resulting from reflection of one or more transmitted signals by one or more first features in the infrastructure of the communication network, the one or more transmitted signals being within the specified frequency range, and (4) capturing, at the second termination device, one or more second reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by the one or more first features in the infrastructure of the communication network. [0142] (C2) The method denoted as (C1) may further include sending the one or more transmitted signals into the infrastructure of the communication network at one or more times that are no later than the specified time period. [0143] (C3) In the method denoted as (C2), the one or more transmitted signals may include at least one of (i) a pilot signal, (ii) a ranging signal, (iii) a sounding signal, and (iv) communication traffic bearing signals. [0144] (C4) Any one of the methods denoted as (C1) through (C3) may further include determining one or more characteristics of the one or more first features in the infrastructure of the communication network at least partially based one or more of (i) the one or more first reflection signals captured by the first termination device from the infrastructure of the communication network and (ii) the one or more second reflection signals captured by the second termination device from the infrastructure of the communication network. [0145] (C5) Any one of the methods denoted as (C1) through (C4) may further include (1) sending the one or more transmitted signals into the infrastructure of the communication network from the first termination device and (2) determining respective locations of the one or more first features in the infrastructure of the communication network relative to the first termination device at least partially based on one or more differences in time between (i) respective times that the one or more first reflection signals are captured by the first termination device and (ii) respective times that the one or more transmitted signals are sent into the communication network from the first termination device. [0146] (C6) Any one of the methods denoted as (C1) through (C5) may further include comparing (i) the one or more first reflection signals captured by the first termination device from the infrastructure of the communication network and (ii) the one or more second reflection signals captured by the second termination device from the infrastructure of the communication network, to determine that the one or more first features in the infrastructure of the communication network are within a communication path of the communication network that is shared by each of the first termination device and the second termination device. [0147] (C7) Any one of the methods denoted as (C1) through (C6) may further include using a ranging and sounding process to determine that the second termination device is within a communication range of the first termination device. [0148] (C8) In any one of the methods denoted as (C1) through (C7), (1) the communication network may be a cable communication network and (2) the first termination device may be a cable modem connected to a coaxial electrical cable of the communication network. [0149] (C9) In any one of the methods denoted as (C1) through (C7), (1) the communication network may be an optical communication network and (2) the first termination device may be one of an optical network termination (ONT) and an optical network unit (ONU) connected to an optical cable of the communication network. [0150] (C10) In any one of the methods denoted as (C1) through (C7), the communication network may be a coherent optical communication network. [0151] (C11) In any one of the methods denoted as (C1) through (C7), (1) the communication network may be a wireless communication network and (2) the first termination device may be a wireless modem. [0152] (D1) A method for determining respective locations of one or more features of a communication network relative to a cable modem of the communication network includes (1) controlling the cable modem and a network hub of the communication network to determine a channel response of the communication network between the cable modem and the network hub using one or more of (i) a channel estimation process, (ii) a pre-equalization process, and (iii) a symbol capture process, (2) converting the channel response from a frequency domain to a time domain, (3) determining signal phase information from the channel response in the time domain, and (4) estimating respective distances of the one or more features of the communication network from the cable modem at least partially based on the signal phase information. [0153] (D2) In the method denoted as (D1), the network hub may include a cable modem termination system. [0154] (D3) In any one of the methods denoted as (D1) and (D2), the network hub may include a remote physical layer (R-PHY) device. [0155] (D4) In any one of the methods denoted as (D1) through (D3), the one or more features of the communication network may include one or more impairments in infrastructure of the communication network.

[0156] Changes may be made in the above methods, devices, and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

Claims

1. A method for identifying a feature in a communication network, the method comprising: receiving, at a first termination device of the communication network, a message specifying an assignment of first network resources of the communication network to the first termination device; sending, from the first termination device, one or more transmitted signals into infrastructure of the communication network at one or more times specified by the assignment of the first network resources, the one or more transmitted signals being within one or more frequency ranges specified by the assignment of the first network resources; and capturing, at the first termination device, one or more reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by one or more features in the infrastructure of the communication network.

2. The method of claim 1, wherein the one or more transmitted signals comprises one or more pilot signals.

3. The method of claim 1, wherein the one or more transmitted signals comprise at least one of (i) a ranging signal and (ii) a sounding signal.

4. The method of claim 1, wherein the one or more times specified by the assignment of the first network resources correspond to one or more quiet periods of the communication network.

5. The method of claim 1, wherein the first network resources comprise one or more time periods and one or more frequency ranges.

6. The method of claim 1, further comprising determining one or more characteristics of the one or more features in the infrastructure of the communication network at least partially based on the one or more reflection signals captured by the first termination device.

7. The method of claim 1, further comprising determining respective locations of the one or more features in the infrastructure of the communication network relative to the first termination device at least partially based on one or more differences in time between (i) respective times that the one or more reflection signals are captured by the first termination device and (ii) respective times that the one or more transmitted signals are sent into the infrastructure of the communication network from the first termination device.

8. The method of claim 1, further comprising determining respective reflection strengths of the one or more features in the infrastructure of the communication network at least partially based on the one or more reflection signals captured by the first termination device from the infrastructure of the communication network.

9. The method of claim 1, further comprising determining presence of a non-linear feature in response to the one or more reflection signals captured by the first termination device from the infrastructure of the communication network being within a frequency range that is different from a frequency range of the one or more transmitted signals.

10. The method of claim 1, further comprising determining presence of signal ingress in response to one or more additional signals captured by the first termination device from the infrastructure of the communication network being within a frequency range that is different from a frequency range of the one or more transmitted signals.

11. The method of claim 1, wherein the first termination device is capable of receiving communication signals having respective frequencies of the one or more transmitted signals.

12. The method of claim 1, wherein: the communication network is a cable communication network; and the first termination device is a cable modem connected to a coaxial electrical cable of the communication network.
13. A method for identifying a feature in a communication network, the method comprising: receiving, at a first termination device of the communication network, a message instructing the first termination device to monitor infrastructure of the communication network for presence of signals within a specified time period and within a specified frequency range; capturing, at the first termination device, a transmitted signal during the specified time period; sending, from the first termination device, a loopback signal into the infrastructure of the communication network, the loopback signal being a function of the transmitted signal captured by the first termination device; and capturing, at the first termination device, one or more first reflection signals from the infrastructure of the communication network resulting from reflection of the loopback signal by one or more features of the infrastructure of the communication network.
14. The method of claim 13, further comprising sending, from a second termination device of the communication network, the transmitted signal into the infrastructure of the communication network.
15. The method of claim 13, further comprising determining respective locations of the one or more features in the infrastructure of the communication network relative to the first termination device at least partially based on one or more differences in time between (i) respective times that the one or more first reflection signals are captured by the first termination device and (ii) respective times that the loopback signal is sent into the infrastructure of the communication network from the first termination device.
16. The method of claim 13, further comprising determining respective locations of the one or more features in the infrastructure of the communication network relative to a second termination device of the communication network at least partially based on differences between (i) distance between the second termination device and the first termination device and (ii) respective distances of the one or more features in the infrastructure of the communication network from the first termination device.
17. The method of claim 13, wherein: the communication network is a cable communication network; and the first termination device is a cable modem connected to a coaxial electrical cable of the communication network.
18. A method for identifying a feature in a communication network, the method comprising: receiving, at a first termination device of the communication network, a message instructing the first termination device to monitor infrastructure of the communication network for presence of signals within a specified time period and within a specified frequency range; receiving, at a second termination device of the communication network, a message instructing the second termination device to monitor the infrastructure of the communication network for presence of signals within the specified time period and within the specified frequency range; capturing, at the first termination device, one or more first reflection signals from the infrastructure of the communication network resulting from reflection of one or more transmitted signals by one or more first features in the infrastructure of the communication network, the one or more transmitted signals being within the specified frequency range; and capturing, at the second termination device, one or more second reflection signals from the infrastructure of the communication network resulting from reflection of the one or more transmitted signals by the one or more first features in the infrastructure of the communication network.
19. The method of claim 18, further comprising: sending the one or more transmitted signals into the infrastructure of the communication network from the first termination device; and determining respective locations of the one or more first features in the infrastructure of the communication network relative to the first termination device at least partially based on one or more differences in time between (i) respective times that the one or more first reflection signals are captured by the

first termination device and (ii) respective times that the one or more transmitted signals are sent into the communication network from the first termination device.

20. The method of claim 18, wherein: the communication network is a cable communication network; and the first termination device is a cable modem connected to a coaxial electrical cable of the communication network.